ARKHAM GENERAL HOSPITAL INFRASTRUCTURE DESIGN

1. Project Overview

Arkham General Hospital is a simulated enterprise hospital environment designed for approximately 200 employees operating 24/7.

The project demonstrates the design and administration of:

- Cisco enterprise networking
- VLAN Segmentation
- Layer 3 inter-VLAN routing
- HSRP gateway redundancy
- EtherChannel
- Windows Server 2022
- Active Directory
- DNS
- File Services
- SMB shares
- Security Groups
- Group Policy
- Automated drive mapping
- PowerShell administration
- End-user troubleshooting

The Cisco network is built in Packet Tracer, while the Windows infrastructure is implemented separately using VirtualBox.

2. Network Architecture

The original network design uses the following VLAN structure:

The hospital network was designed using a hierarchical network architecture to provide departmental segmentation, centralized services, redundancy, and scalable connectivity for end-user and server systems.

The architecture separates network infrastructure into distinct functional layers. Layer 3 distribution switches provide inter-VLAN routing and redundant default-gateway services, while access-layer switches provide connectivity for departmental endpoints.

The network also includes a dedicated server environment supporting centralized infrastructure services such as Active Directory, DNS, DHCP, file services, authentication, and Group Policy.

2.1 High-Level Architecture

The network architecture consists of the following major components:

- Internet/ISP connectivity
- Layer 3 distribution switches
- Layer 2 access switches
- Server infrastructure
- End-user workstations
- Departmental VLANs
- Centralized Windows Server services

2.2 Layer 3 Distribution

The Layer 3 distribution switches provide the primary routing functions within the internal network.

Switch Virtual Interfaces (SVIs) are configured for the departmental VLANs and provide the default gateways used by connected hosts.

The distribution layer also uses HSRP to provide redundant default-gateway functionality. This allows hosts to continue using a virtual gateway address if one distribution switch becomes unavailable.

2.3 Layer 2 Access

Access switches provide connectivity between end-user devices and the distribution layer.

VLAN assignments are used to logically separate hospital departments and infrastructure services. Trunk links are used where multiple VLANs must traverse a single physical connection.

EtherChannel is also used to logically aggregate multiple physical links, providing increased link capacity and resilience.

2.4 Server Infrastructure

The server environment provides centralized infrastructure and identity services for the network.

Key services include:

- Active Directory Domain Services
- DNS
- DHCP
- File Services
- Authentication
- Group Policy

The Windows environment uses the arkham.local Active Directory domain.

The file server, FILE01, provides departmental network shares that are accessed by authorized users through SMB.

The Cisco networking environment and Windows server environment were implemented separately. The Cisco infrastructure was simulated using Cisco Packet Tracer, while the Windows infrastructure was implemented using VirtualBox.

2.5 Network Segmentation

Departmental and infrastructure traffic is separated using VLANs. The current network plan includes dedicated VLANs for:

- Administration
- IT
- Clinical Services
- Emergency
- Research
- Server Farm
- Management
- Guest Wi-Fi

Each VLAN is assigned its own IPv4 subnet and Layer 3 default gateway.

3. Network Design Goals

The network was designed around several requirements:

- Segmentation

Departments are separated into individual VLANs to reduce broadcast domains and provide logical network segmentations.

- High Availability

The Layer 3 distribution switches use HSRP to provide a redundant default gateway.

- Link Redundancy

EtherChannel provides logical aggregation of multiple physical links, increasing available bandwidth and providing resilience in case an individual member link fails.

- Centralized Services

The server environment provides:

* Active Directory
* DNS
* DHCP
* File services
* Authentication
* Group Policy

4. Network Topology

The network uses VLAN-based segmentation to separate departmental, server, management, and guest traffic. Each VLAN is assigned a dedicated /24 IPv4 subnet with the Layer 3 distribution switches providing the default gateway.

VLAN	**DEPARTMENT**	**NETWORK**	**DEFAULT GATEWAY**

10	Administration	192.168.10.0/24	192.168.10.1
20	IT	192.168.20.0/24	192.168.20.1
30	Clinical Services	192.168.30.0/24	192.168.30.1
40	Emergency	192.168.40.0/24	192.168.40.1
50	Research	192.168.50.0/24	192.168.50.1
60	Server Farm	192.168.60.0/24	192.168.60.1
70	Management	192.168.70.0/24	192.168.70.1
80	Guest Wi-Fi	192.168.80.0/24	192.168.80.1

5. Layer 3 Routing and High Availability

5.1 Inter-VLAN Routing

Inter-VLAN routing is performed by the Layer 3 distribution switches. Each departmental VLAN is configured with a Switch Virtual Interface (SVI), which provides the Layer 3 gateway for devices within that VLAN.

The SVIs allow traffic to be routed between departmental networks while maintaining logical separation between VLANs.

Each VLAN uses the .1 address as its default gateway:

- VLAN 10 — 192.168.10.1
- VLAN 20 — 192.168.20.1
- VLAN 30 — 192.168.30.1
- VLAN 40 — 192.168.40.1
- VLAN 50 — 192.168.50.1
- VLAN 60 — 192.168.60.1
- VLAN 70 — 192.168.70.1
- VLAN 80 — 192.168.80.1

The distribution switches perform routing between these networks, allowing authorized traffic to move between departments and centralized services.

5.2 SVI Configuration

SVIs were configured on the Layer 3 distribution switches for each required VLAN. The SVIs provide the default gateway used by end devices and servers.

The following command was used to verify SVI status:

Show ip interface brief

This verification confirms that the VLAN interfaces are configured with the appropriate IP addresses and are operational.

DSW1 SVIs	DSW2 SVIs
Vlan10 192.168.10.2 YES manual up up Vlan20 192.168.20.2 YES manual up up Vlan30 192.168.30.2 YES manual up up Vlan40 192.168.40.2 YES manual up up Vlan50 192.168.50.2 YES manual up up Vlan70 192.168.70.2 YES manual up up Vlan80 192.168.80.2 YES manual up up	Vlan10 192.168.10.3 YES manual up up Vlan20 192.168.20.3 YES manual up up Vlan30 192.168.30.3 YES manual up up Vlan40 192.168.40.3 YES manual up up Vlan50 192.168.50.3 YES manual up up Vlan70 192.168.70.3 YES manual up up Vlan80 192.168.80.3 YES manual up up

5.3 HSRP High Availability

Hot Standby Router Protocol (HSRP) was implemented between the Layer 3 distribution switches to provide gateway redundancy.

Rather than relying on a single physical distribution switch as the default gateway, end devices use a virtual IP address provided by HSRP. One distribution switch operates as the active router while the other remains available as the standby router.

If the active distribution switch becomes unavailable, the standby switch can assume the active role, allowing clients to maintain connectivity without manually changing their default gateway.

This design improves network availability and eliminates a single point of failure at the default gateway.

HSRP CONFIGURATION

VLAN	**Active Switch**	**Standby Switch**	**Virtual Gateway**
10	DSW1	DSW2	192.168.10.1
20	DSW2	DSW1	192.168.20.1
30	DSW1	DSW2	192.168.30.1
40	DSW2	DSW1	192.168.40.1
50	DSW1	DSW2	192.168.50.1
60	DSW2	DSW1	192.168.60.1
70	DSW1	DSW2	192.168.70.1
80	DSW2	DSW1	192.168.80.1

5.4 EtherChannel Link Redundancy

EtherChannel was implemented to combine multiple physical links between network devices into a single logical link.

This provides:

- Increased available bandwidth
- Link-level redundancy
- Improved resilience against individual link failures
- Simplified management of multiple physical connections

Etherchannel was verified using:

Show etherchannel summary

DSW1 Etherchannel	DSW2 Etherchannel

`Group  Port-channel  Protocol    Ports`
`------+-------------+-----------+-----------------------------------------------`
`1      Po1(SU)          PAgP   Gig1/0/1(P) Gig1/0/5(P)`
	`Group  Port-channel  Protocol    Ports`
	`------+-------------+-----------+-----------------------------------------------`
	`1      Po1(SU)          PAgP   Gig1/0/1(P) Gig1/0/5(P)`

5.5 Routing Verification

Connectivity between VLANs was tested using ICMP and other network troubleshooting tools.

Examples of verification commands used include:

Show ip route

```
Gateway of last resort is not set

     192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, Vlan10
L       192.168.10.2/32 is directly connected, Vlan10
     192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.20.0/24 is directly connected, Vlan20
L       192.168.20.2/32 is directly connected, Vlan20
     192.168.30.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.30.0/24 is directly connected, Vlan30
L       192.168.30.2/32 is directly connected, Vlan30
     192.168.40.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.40.0/24 is directly connected, Vlan40
L       192.168.40.2/32 is directly connected, Vlan40
     192.168.50.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.50.0/24 is directly connected, Vlan50
L       192.168.50.2/32 is directly connected, Vlan50
     192.168.70.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.70.0/24 is directly connected, Vlan70
L       192.168.70.2/32 is directly connected, Vlan70
     192.168.80.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.80.0/24 is directly connected, Vlan80
L       192.168.80.2/32 is directly connected, Vlan80
```

Show ip interface brief

```
Vlan10                 192.168.10.2    YES manual up                    up
Vlan20                 192.168.20.2    YES manual up                    up
Vlan30                 192.168.30.2    YES manual up                    up
Vlan40                 192.168.40.2    YES manual up                    up
Vlan50                 192.168.50.2    YES manual up                    up
Vlan70                 192.168.70.2    YES manual up                    up
Vlan80                 192.168.80.2    YES manual up                    up
```

Ping <destination-ip>

```
C:\>ping 192.168.20.1

Pinging 192.168.20.1 with 32 bytes of data:

Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.20.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Ping from Admin PC To IT Default Gateway

Traceroute <destination-ip>

```
C:\>tracert 192.168.20.11

Tracing route to 192.168.20.11 over a maximum of 30 hops:

  1   0 ms      0 ms      0 ms      192.168.10.2
  2   0 ms      0 ms      0 ms      192.168.20.11

Trace complete.
```

Traceroute from Admin PC to IT PC

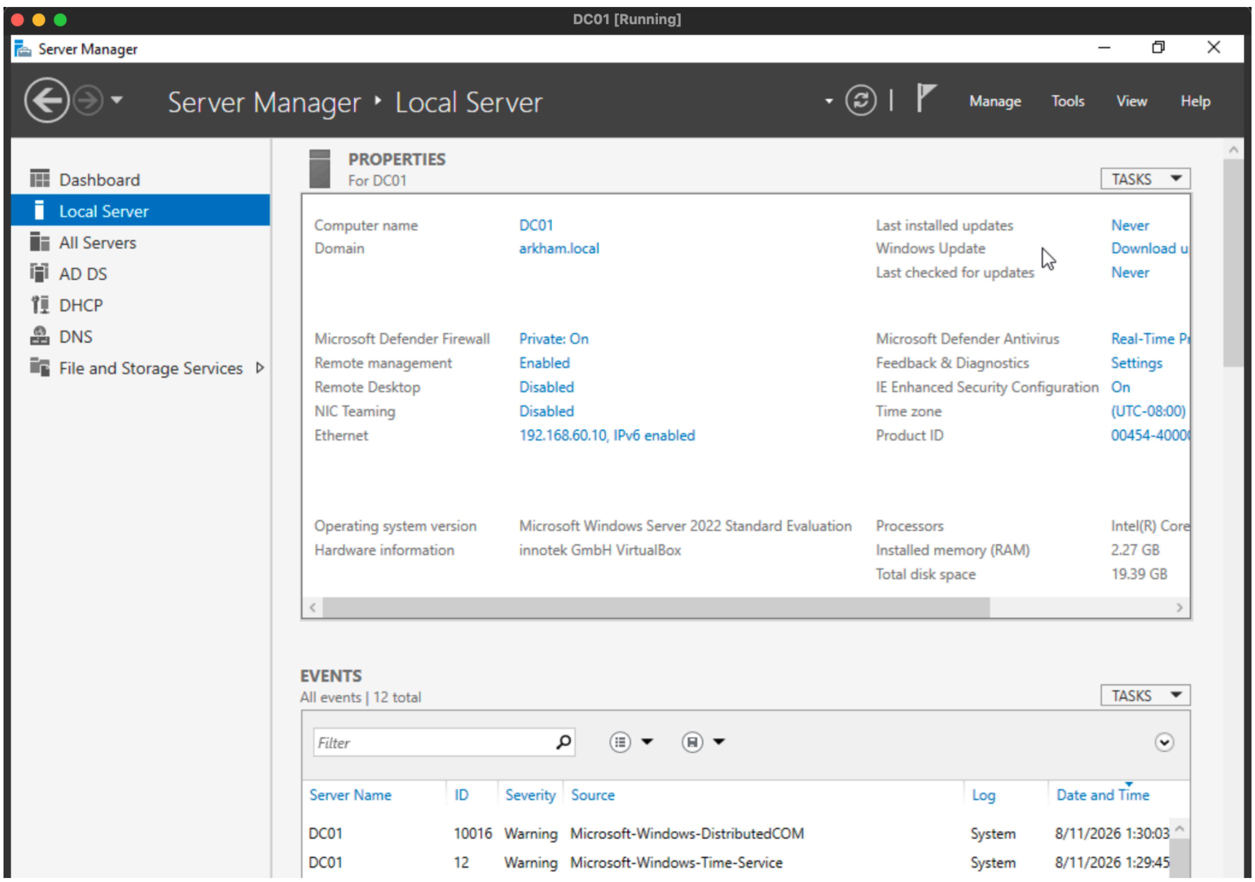
These tests were used to verify that Layer 3 interfaces were operational and that traffic could successfully traverse the network between appropriate VLANs.

VLAN	**Subnet**	**DSW1 SVI**	**DSW2 SVI**	**HSRP Virtual Gateway**
10 - Administration	192.168.10.0	192.168.10.2	192.168.10.3	192.168.10.1
20 - IT	192.168.20.0	192.168.20.2	192.168.20.3	192.168.20.1
30 - Clinical	192.168.30.0	192.168.30.2	192.168.30.3	192.168.30.1
40 - Emergency	192.168.40.0	192.168.40.2	192.168.40.3	192.168.40.1

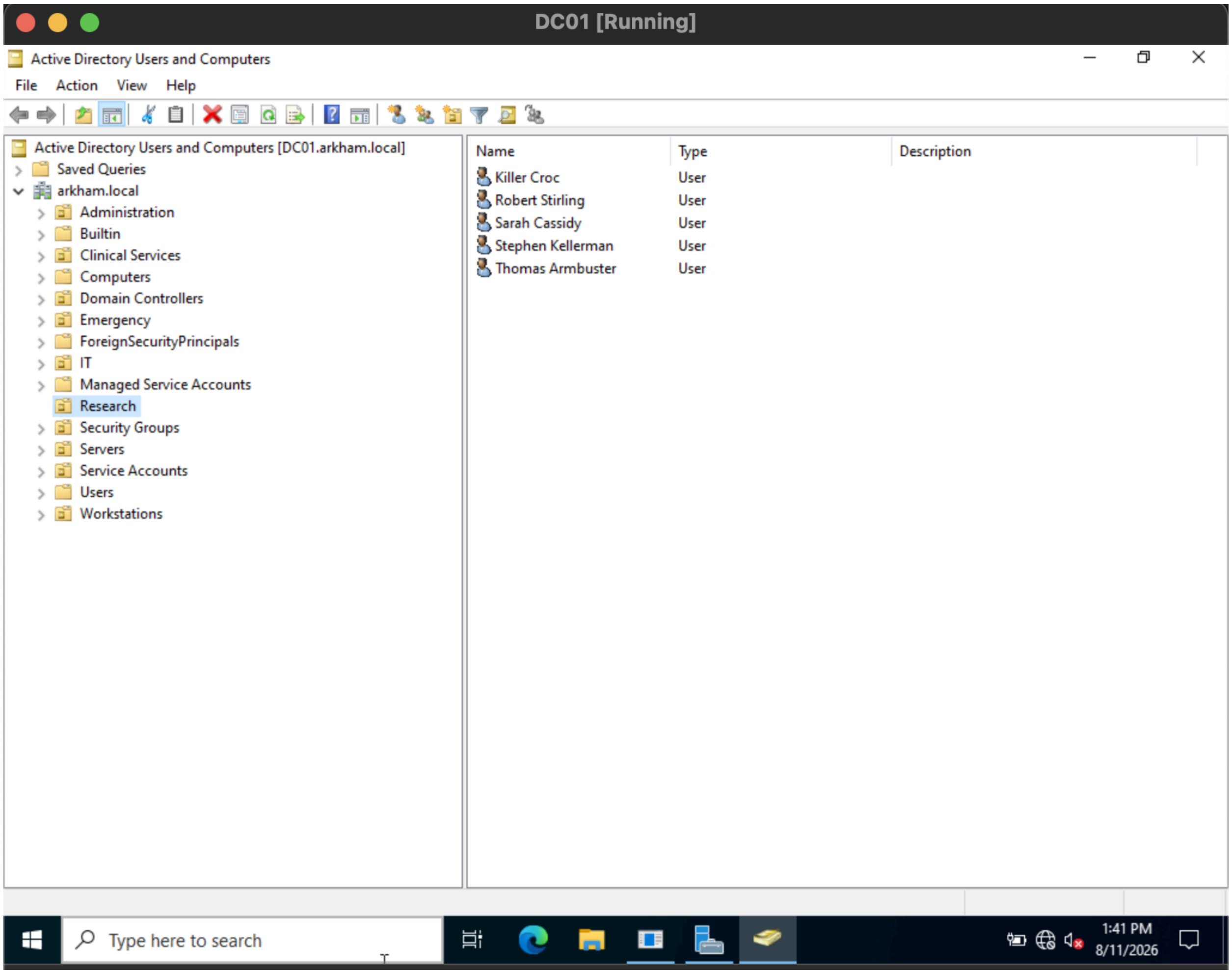
50 - Research	192.168.50.0	192.168.50.2	192.168.50.3	192.168.50.1
60 - Servers	192.168.60.0	192.168.60.2	192.168.60.3	192.168.60.1
70 - Management	192.168.70.0	192.168.70.2	192.168.70.3	192.168.70.1
80 - Guest	192.168.80.0	192.168.80.2	192.168.80.3	192.168.80.1

6. Windows Server Infrastructure

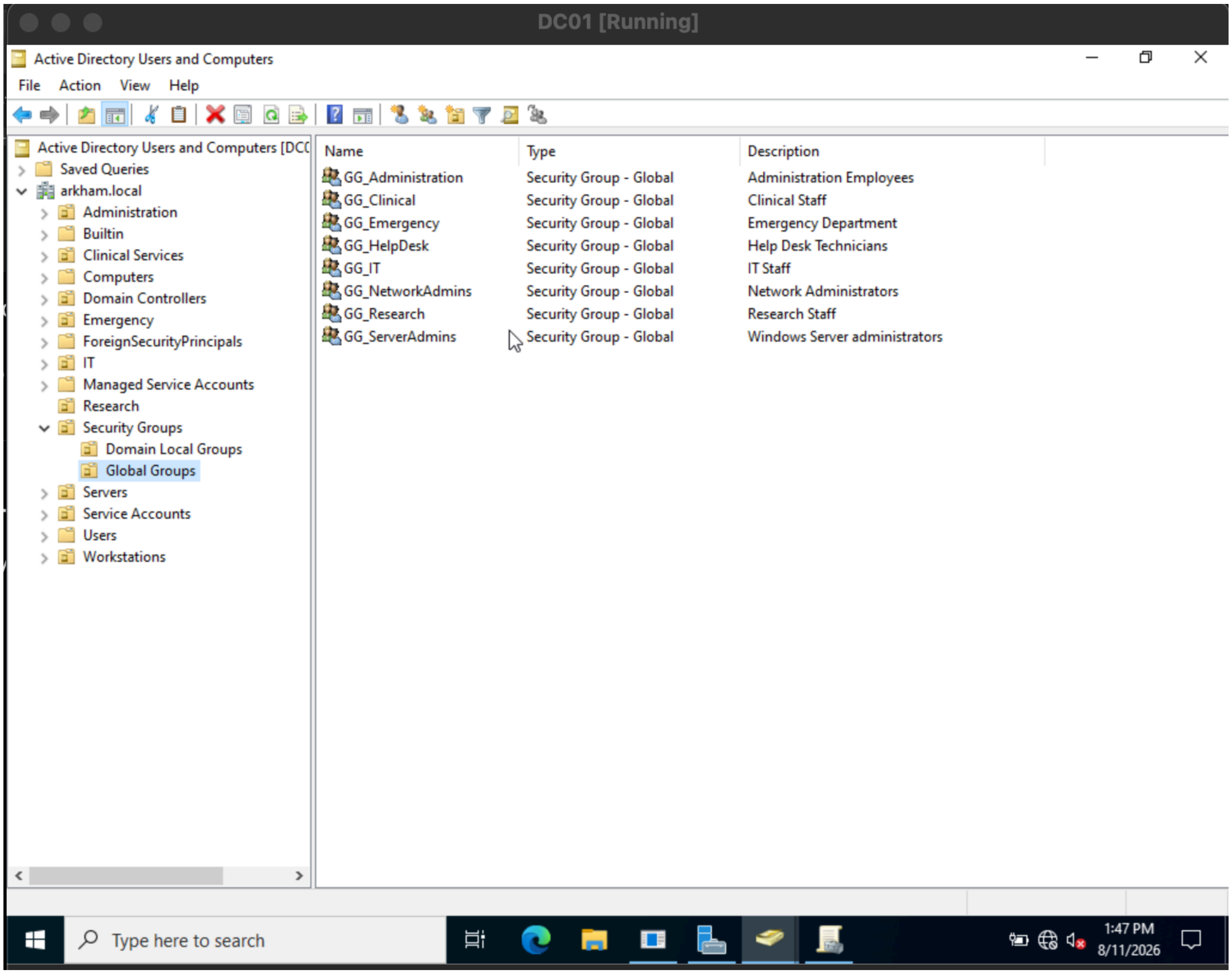
6.1 Active Directory Domain Services



Active Directory Service

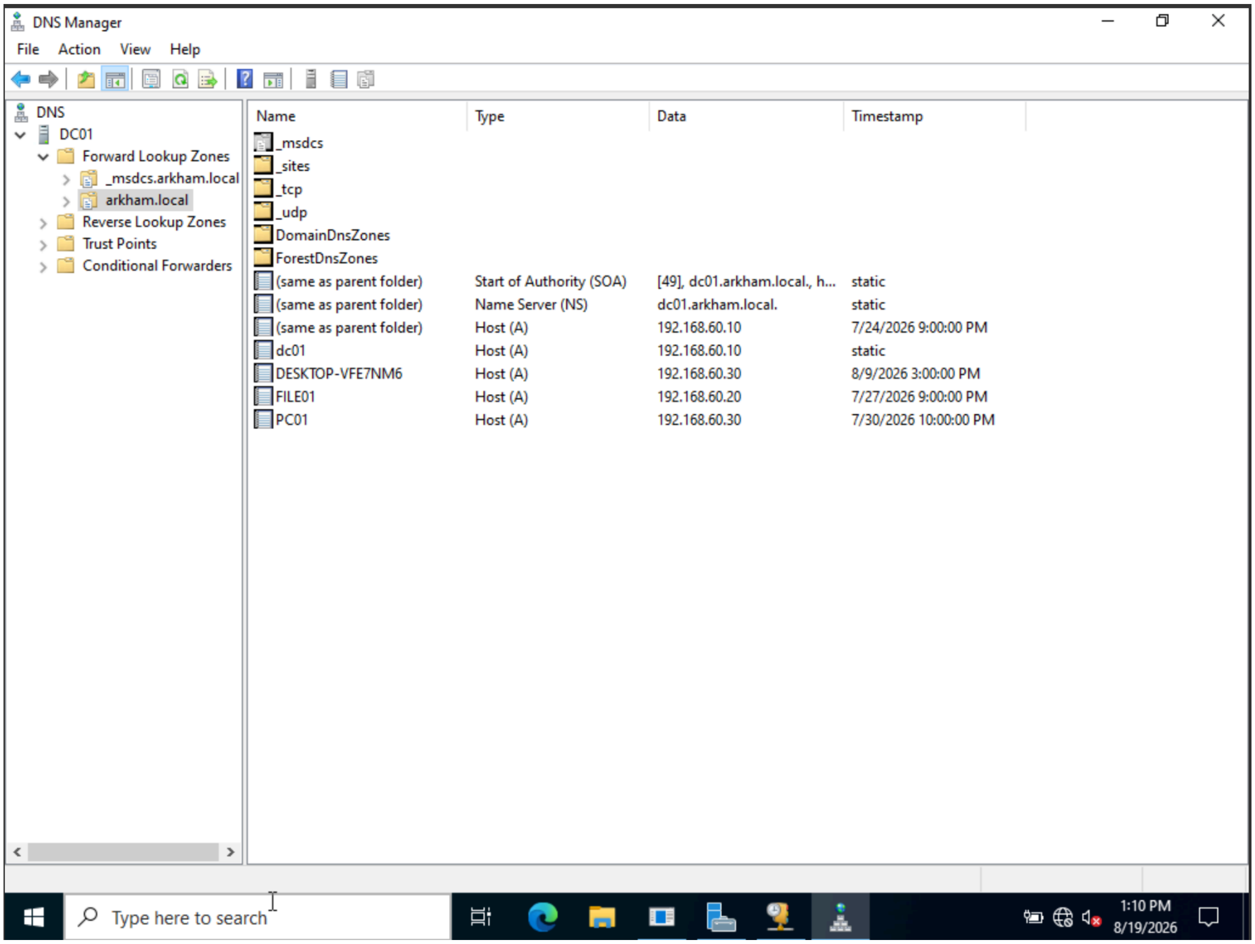


Organizational Units



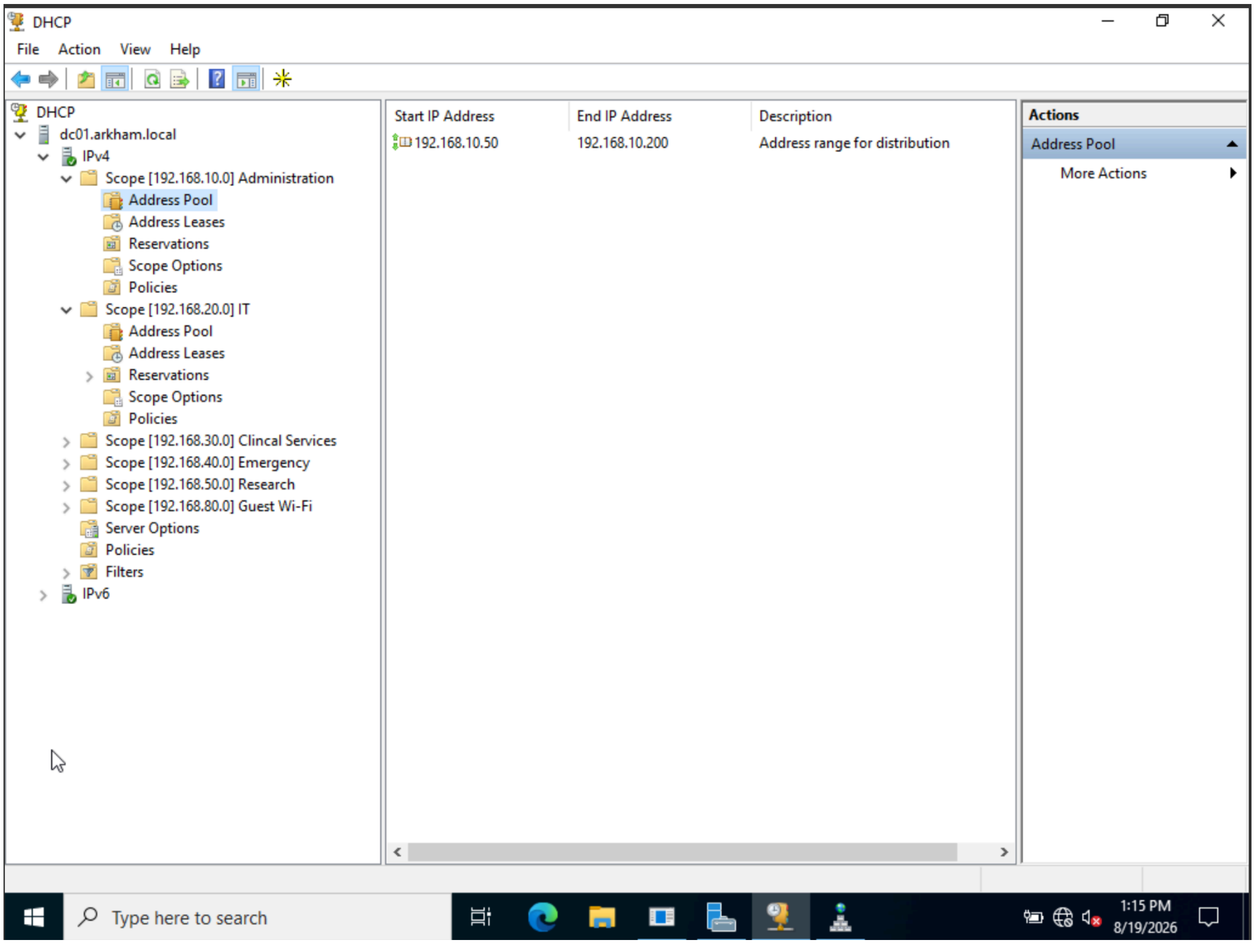
Global Groups

6.2 DNS



DNS Manager showing the arkham.local forward lookup zone and host records for domain infrastructure and network devices.

6.3 DHCP



DHCP Manager showing centralized DHCP scopes configured on DC01 for multiple hospital VLANs.

DHCP Services

DHCP was deployed on DC01 to provide centralized IP address assignment for client devices across multiple VLANs. Separate DHCP scopes were configured for departmental networks, allowing clients to automatically receive appropriate IP addressing based on their VLAN.

The Administration scope uses the address range 192.168.10.50–192.168.10.200, while additional scopes are configured for IT, Clinical Services, Emergency, Research, and Guest Wi-Fi.

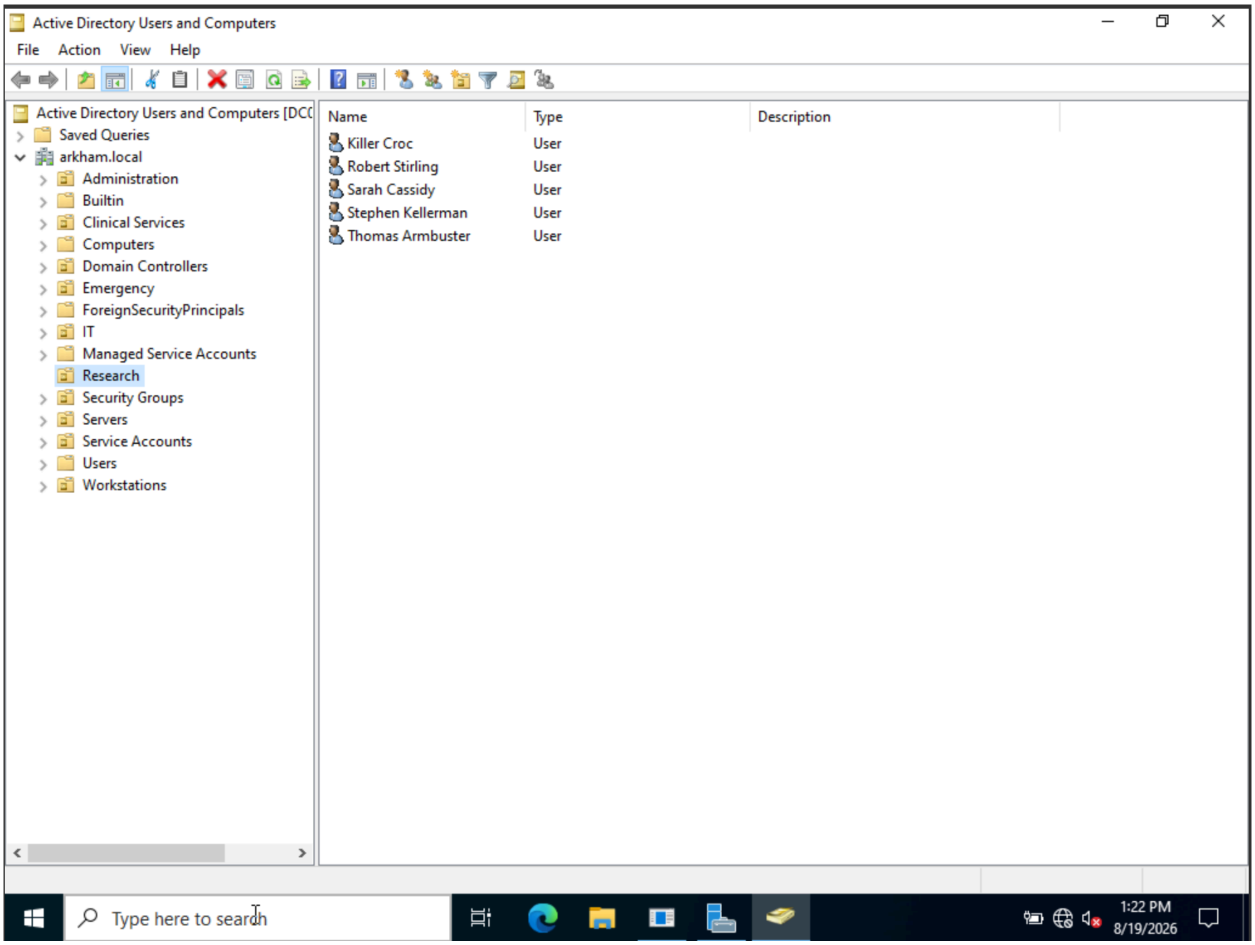
6.4 Active Directory and Role-Based Access Control

Active Directory Domain Services was deployed using the arkham.local domain. Organizational Units were created to separate users, workstations, servers, service accounts, and departmental resources.

Security groups were implemented to provide role-based access control. Global groups represent departmental or administrative roles, while domain local groups define the level of access granted to specific resources.

The file access model separates Read-Only (RO) and Read/Write (RW) permissions for departmental file shares.

Figure 6.4.1 — Active Directory organizational structure



Active Directory Users and computer showing the arkham.local domain structure and departmental organizational units.

Figure 6.4.2 — Domain local resource-access groups

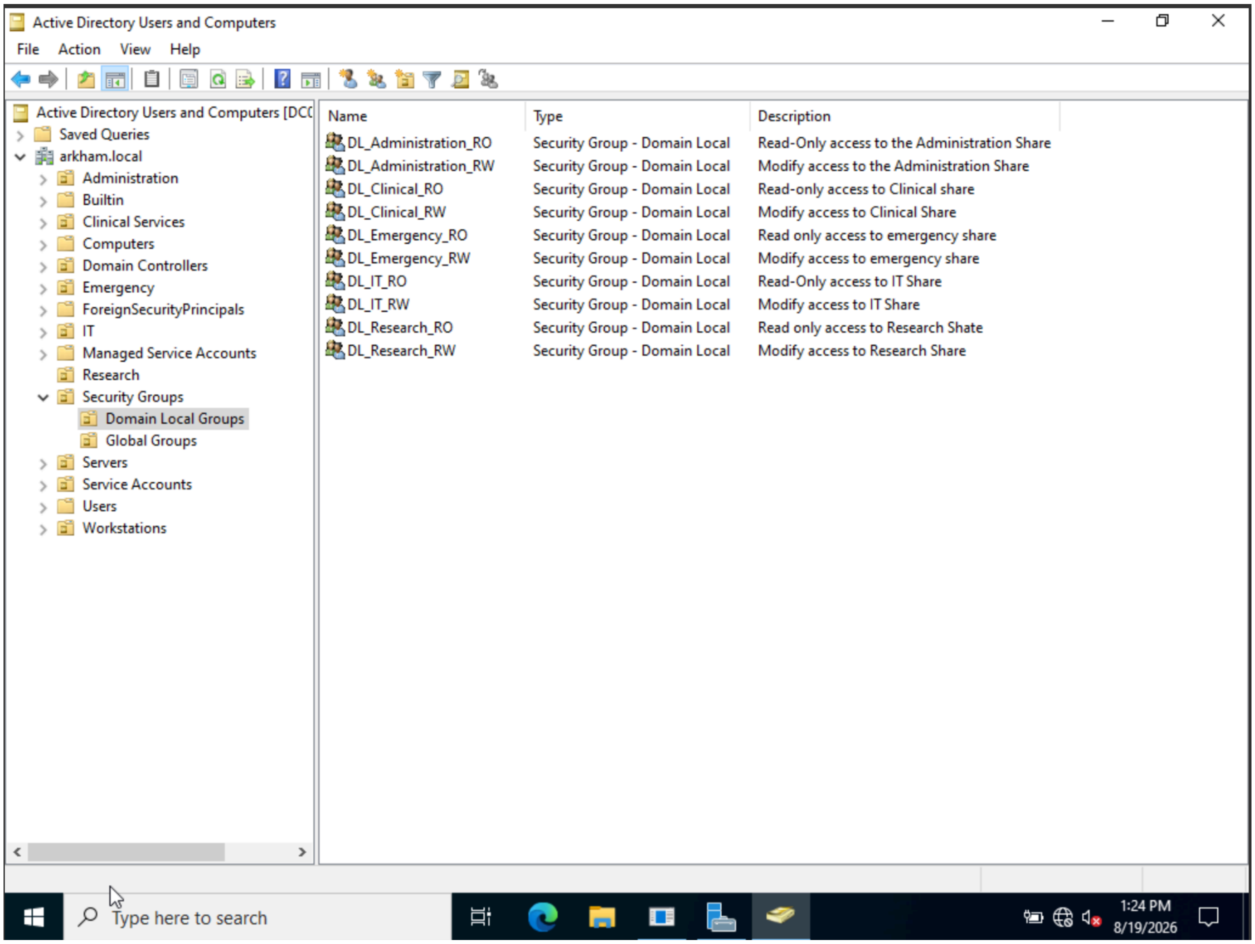


Figure 6.4.3 — Global security groups

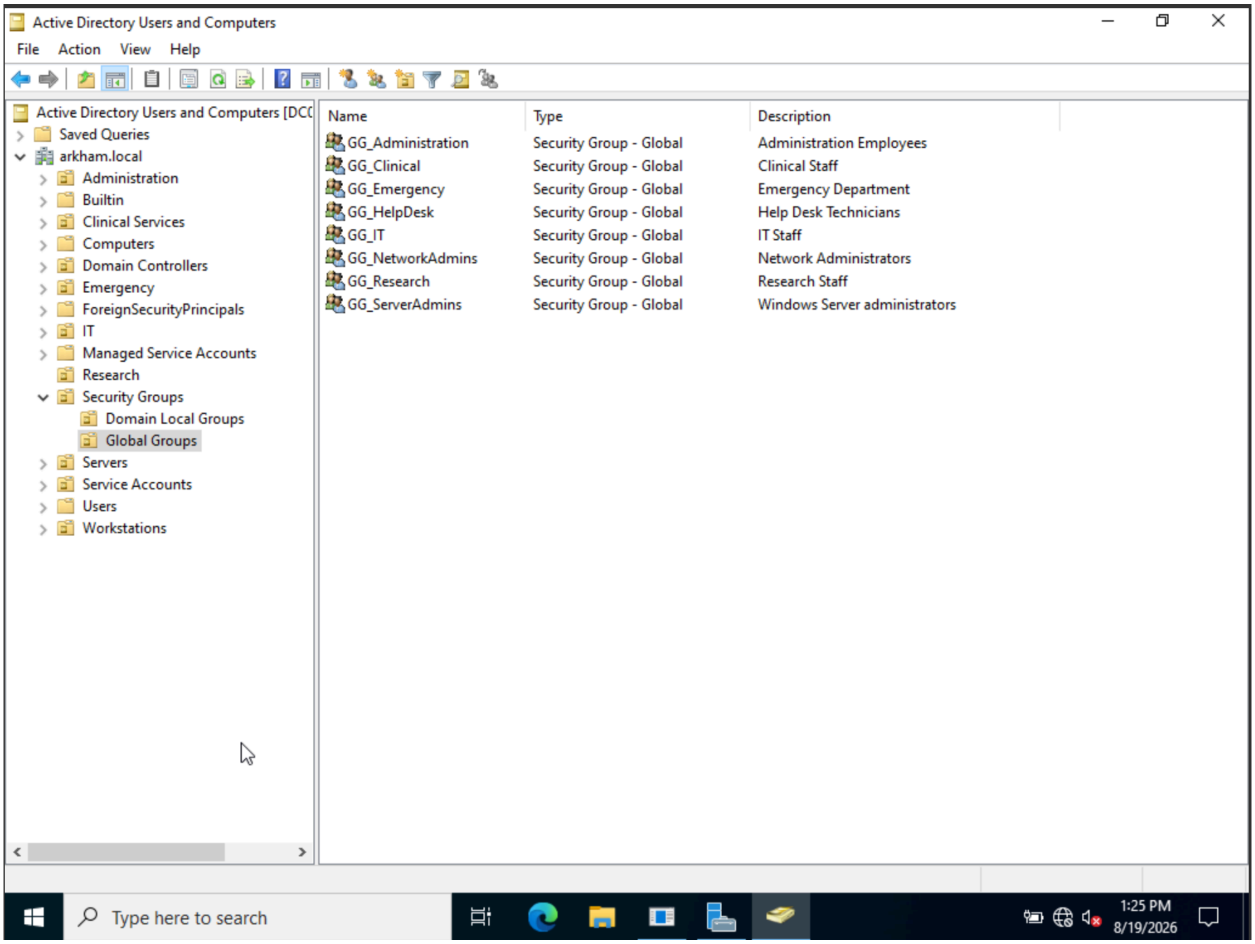


Figure 6.4.4 - Bruce Wayne's Active Directory group membership showing assignment to the Administration security group

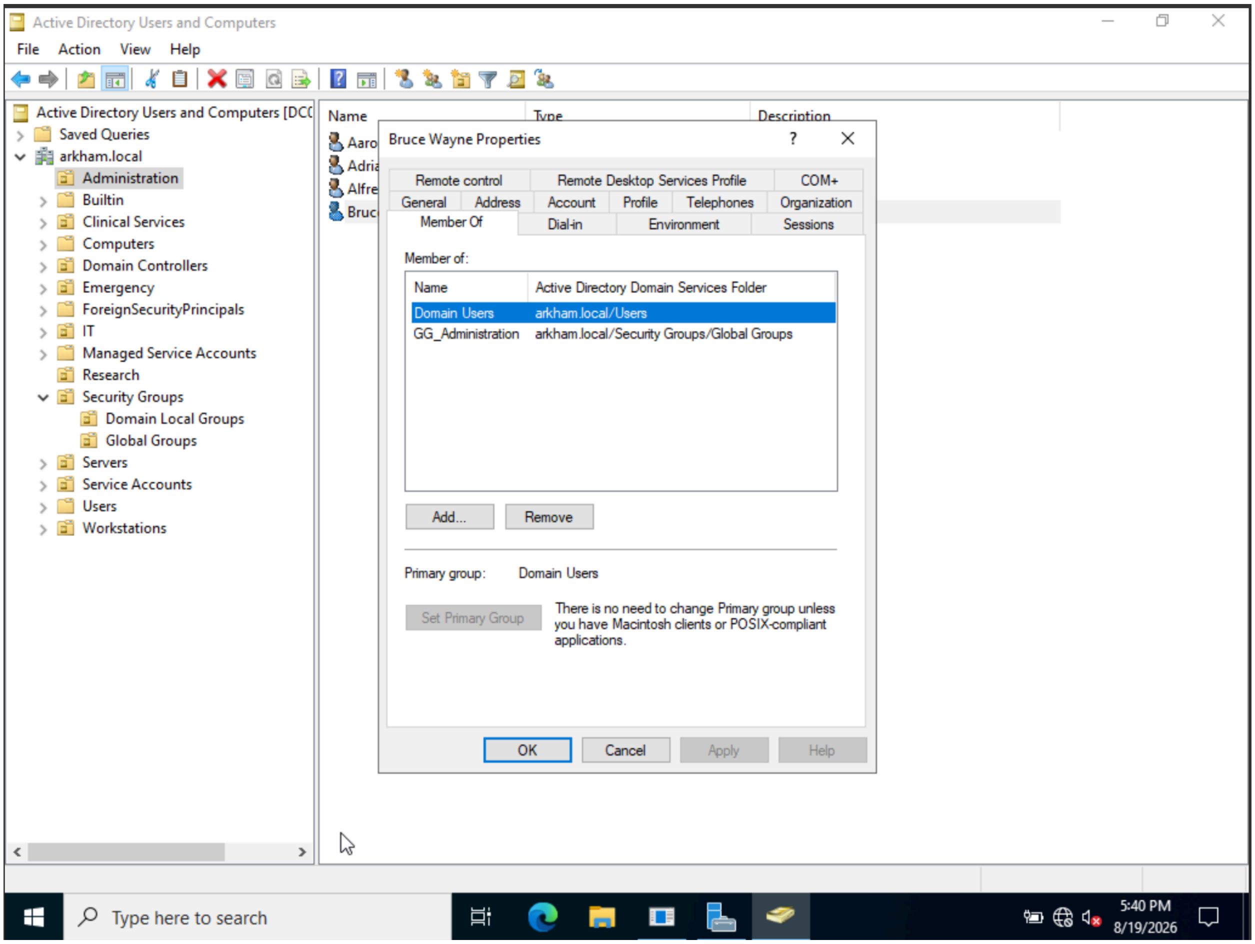
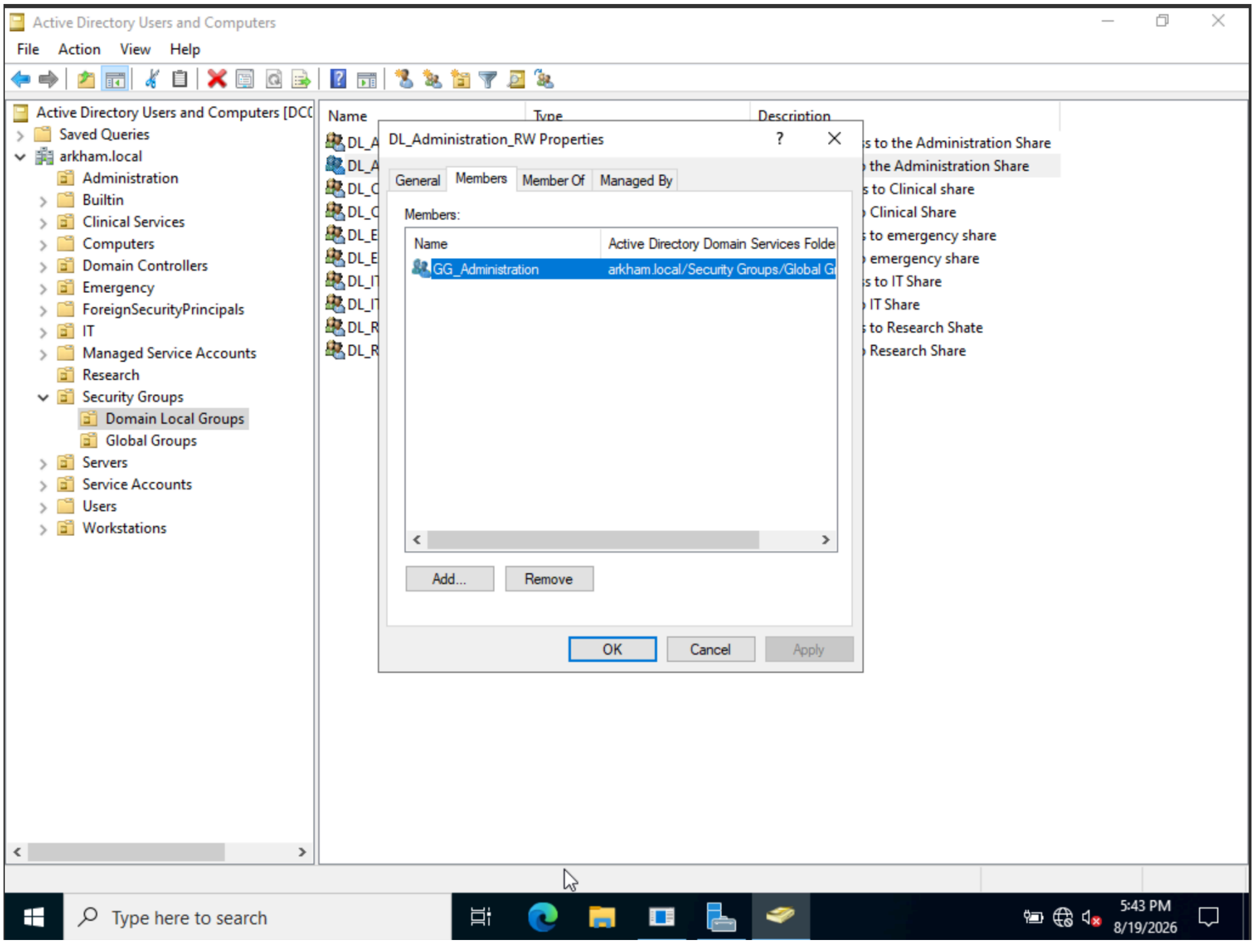


Figure 6.4.5 - Domain Local security group membership showing GG_Administration nested within DL_Administration_RW



<u>6.5 Group Policy and Automated Drive Mapping</u>

Figure 6.5.1 - Clinical department network drive mapping configured through Group Policy Preferences

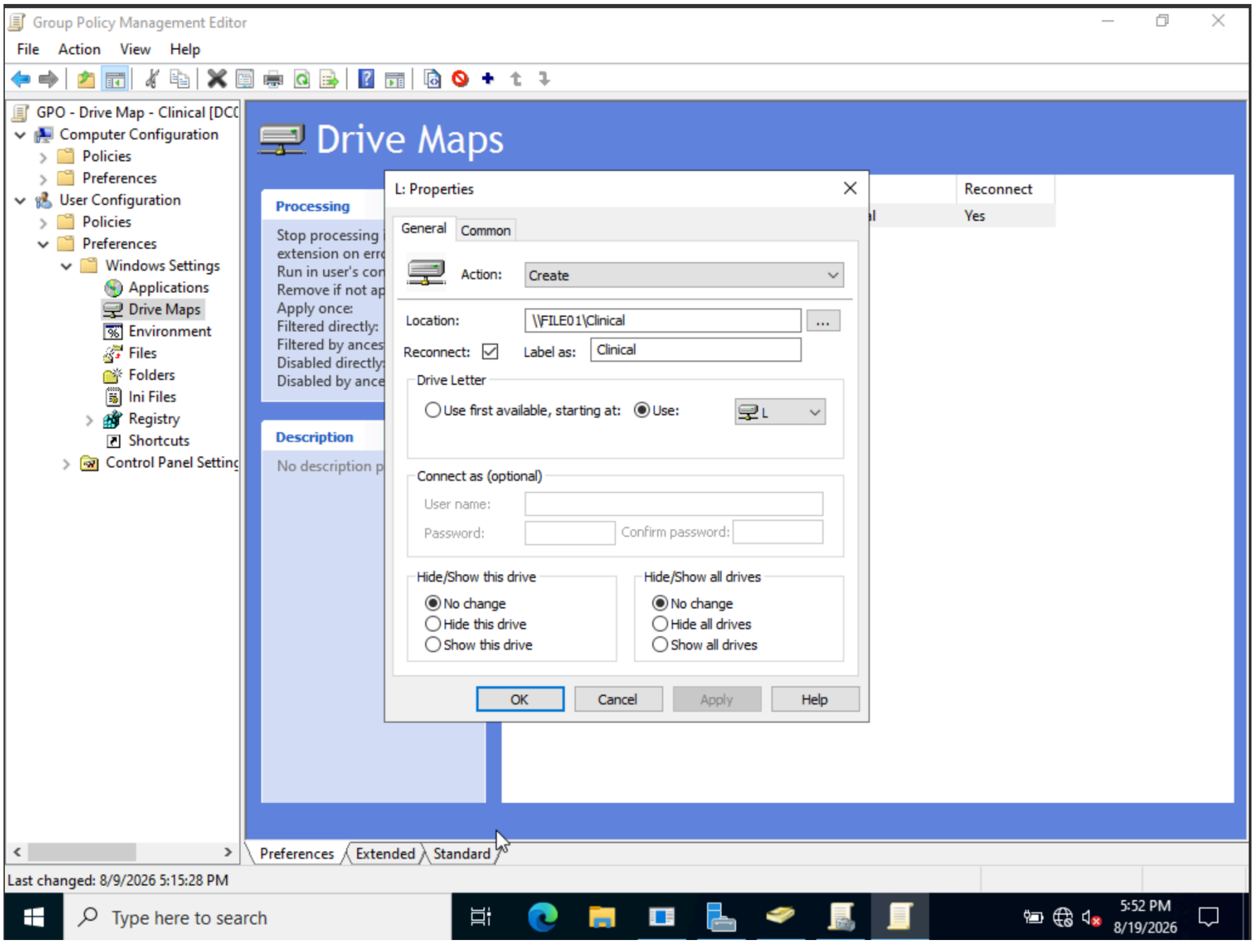


Figure 6.5.2 - Item-level targeting restricts the Clinical drive mapping to members of the GG_Clinical security group

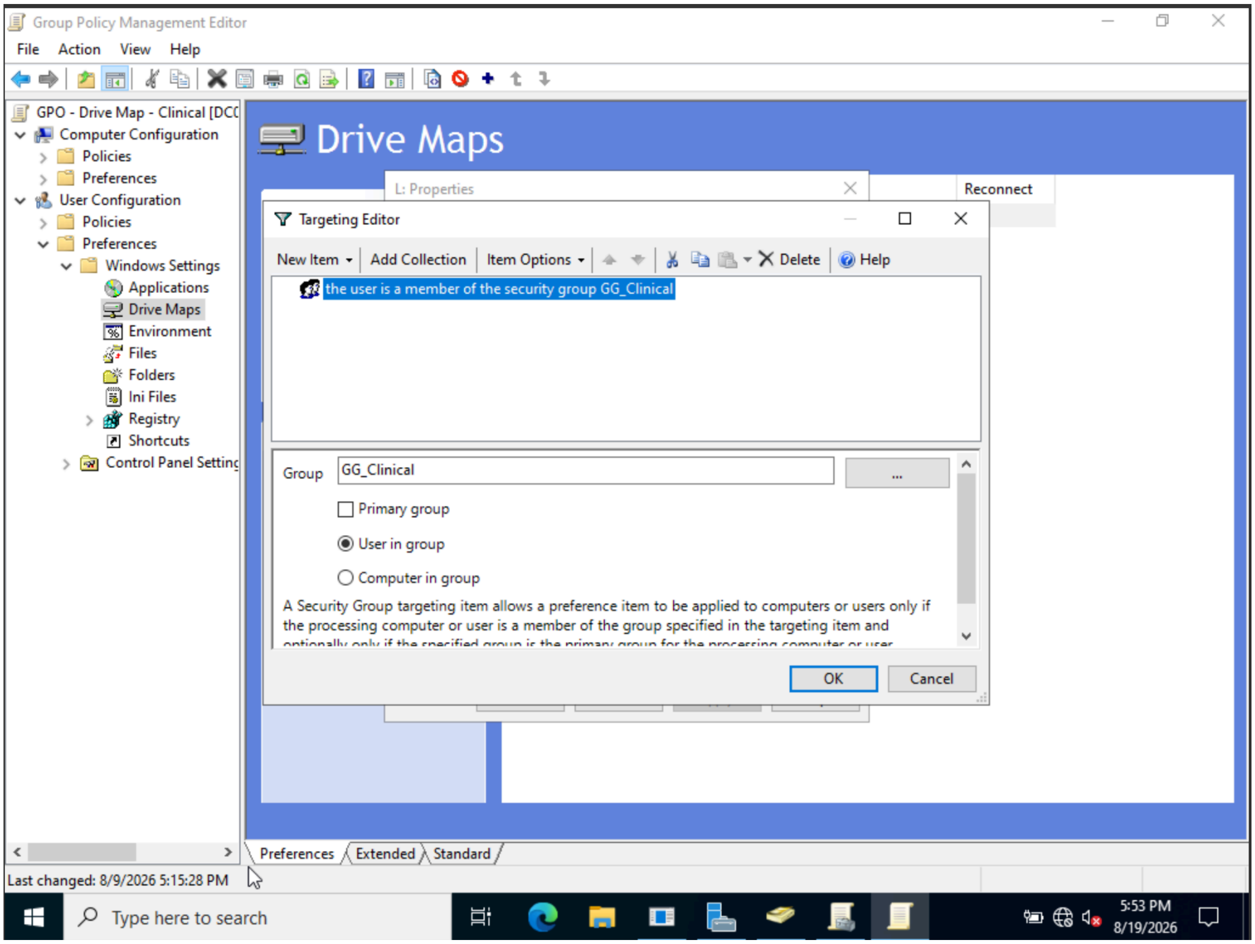


Figure 6.5.3 - The Clinical drive-mapping GPO is linked to the Clinical Services OU

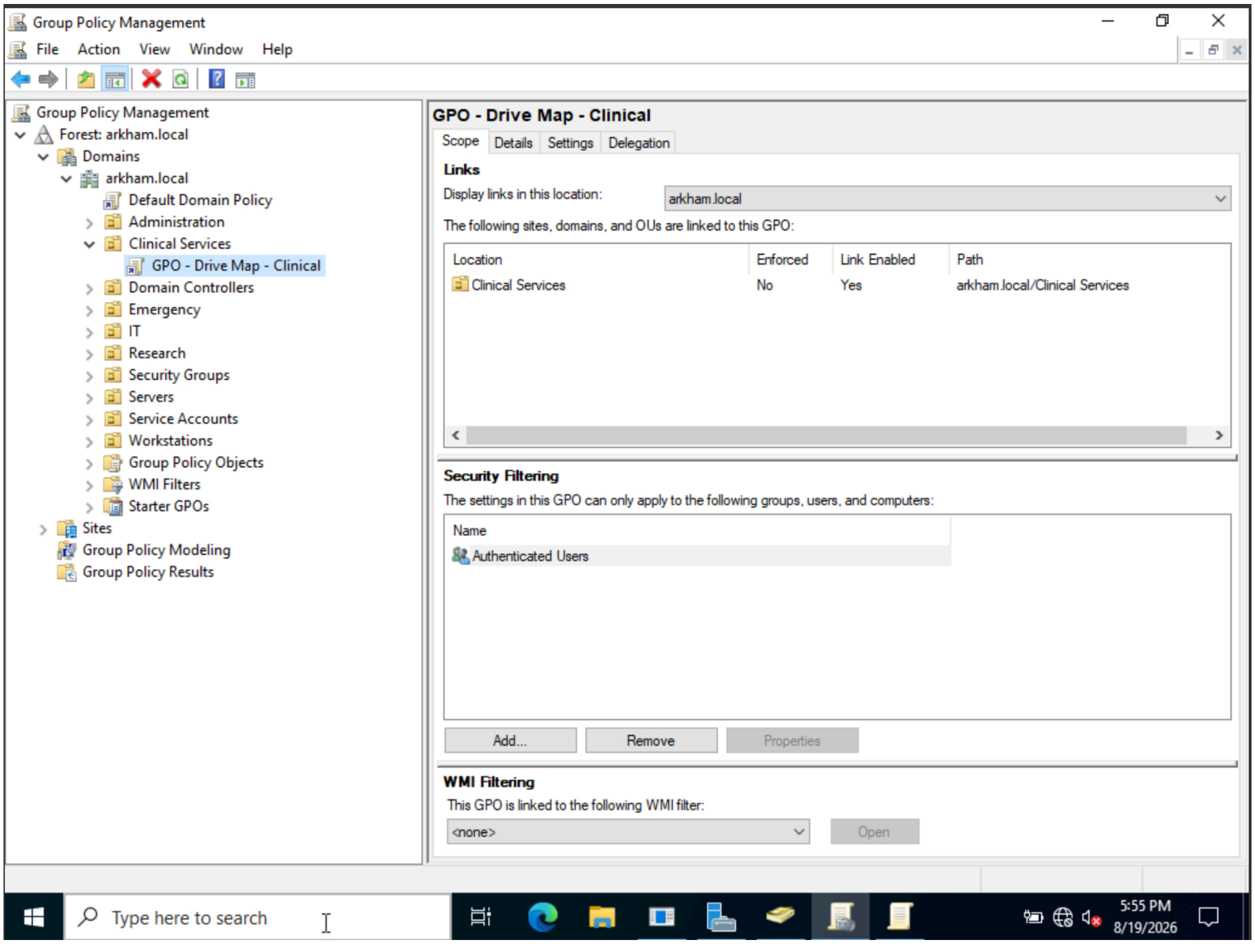
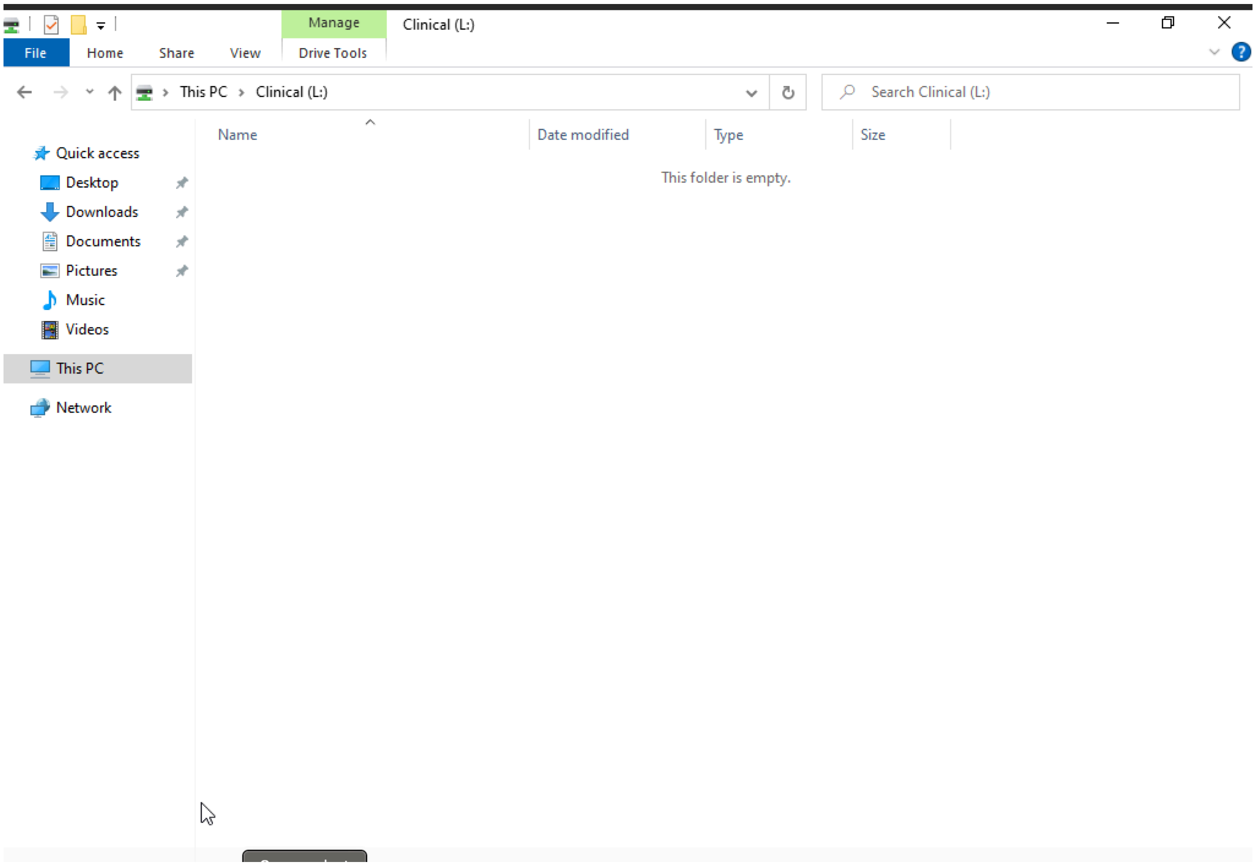


Figure 6.4 - Successful client-side validation of the Clinical network drive mapping

Command Prompt

```
OS Configuration:            Member Workstation
OS Version:                  10.0.19045
Site Name:                   N/A
Roaming Profile:             N/A
Local Profile:               C:\Users\jfraser
Connected over a slow link?: No


USER SETTINGS
--------------
    CN=Jordan Fraser,OU=Clinical Services,DC=arkham,DC=local
    Last time Group Policy was applied: 8/19/2026 at 6:16:23 PM
    Group Policy was applied from:      DC01.arkham.local
    Group Policy slow link threshold:   500 kbps
    Domain Name:                        ARKHAM
    Domain Type:                        Windows 2008 or later

    Applied Group Policy Objects
    -----------------------------
        GPO - Drive Map - Clinical

    The following GPOs were not applied because they were filtered out
    -------------------------------------------------------------------
        Local Group Policy
            Filtering:  Not Applied (Empty)

    The user is a part of the following security groups
    ---------------------------------------------------
        Domain Users
        Everyone
        BUILTIN\Users
        NT AUTHORITY\INTERACTIVE
        CONSOLE LOGON
        NT AUTHORITY\Authenticated Users
        This Organization
        LOCAL
        GG_Clinical
        Authentication authority asserted identity
        DL_Clinical_RW
        Medium Mandatory Level

C:\Users\jfraser>
```



The Clinical file share was automatically mapped to drive L: using Group Policy Preferences. Item-level targeting was configured to restrict the mapping to members of the GG_Clinical security group. The policy was linked to the Clinical Services OU and validated using gpresult /r. Successful client-side validation confirmed that the Clinical share was automatically presented as drive L: and was accessible to the authenticated user.

6.7 PowerShell Administration

Figure 6.7.1 - Powershell was used to filter users by their names and User IDs

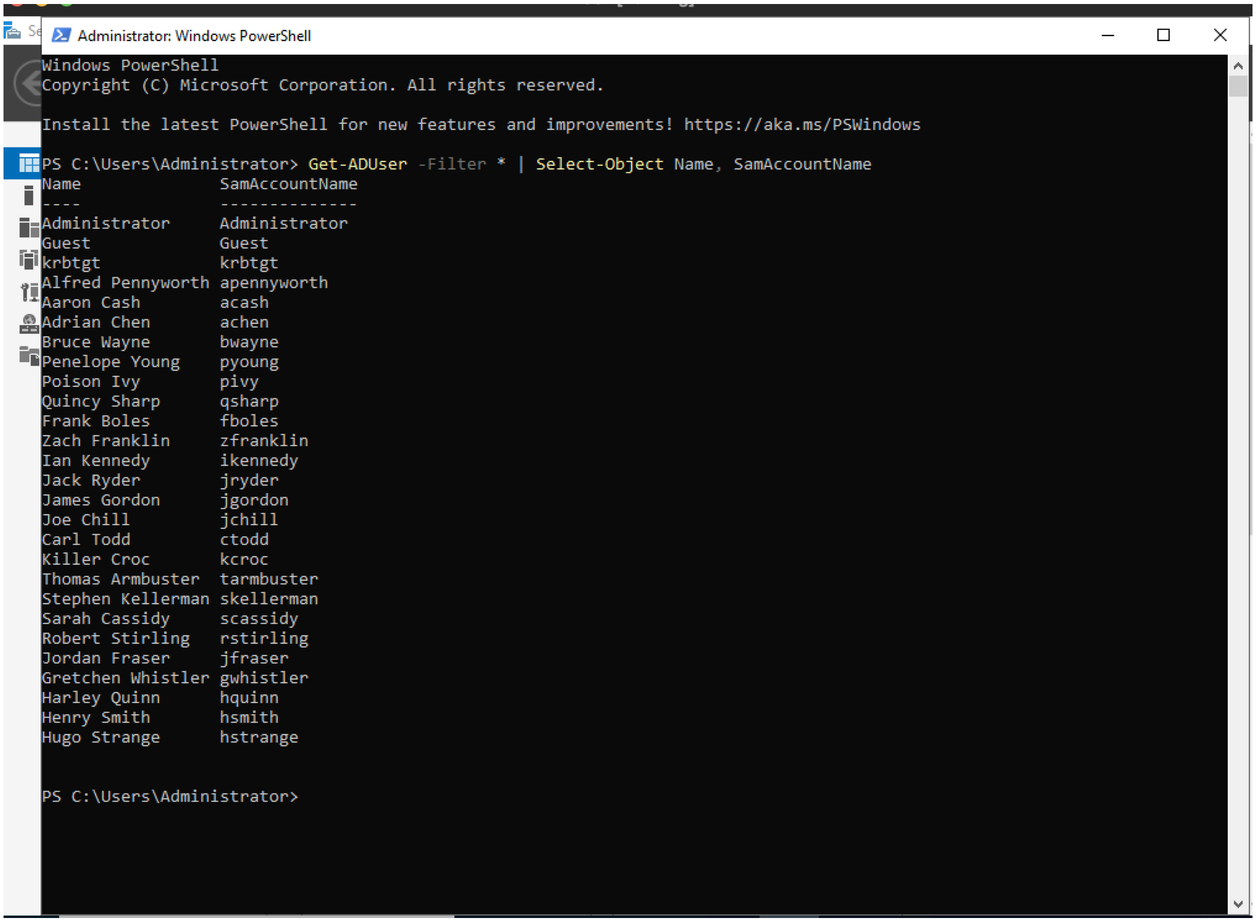
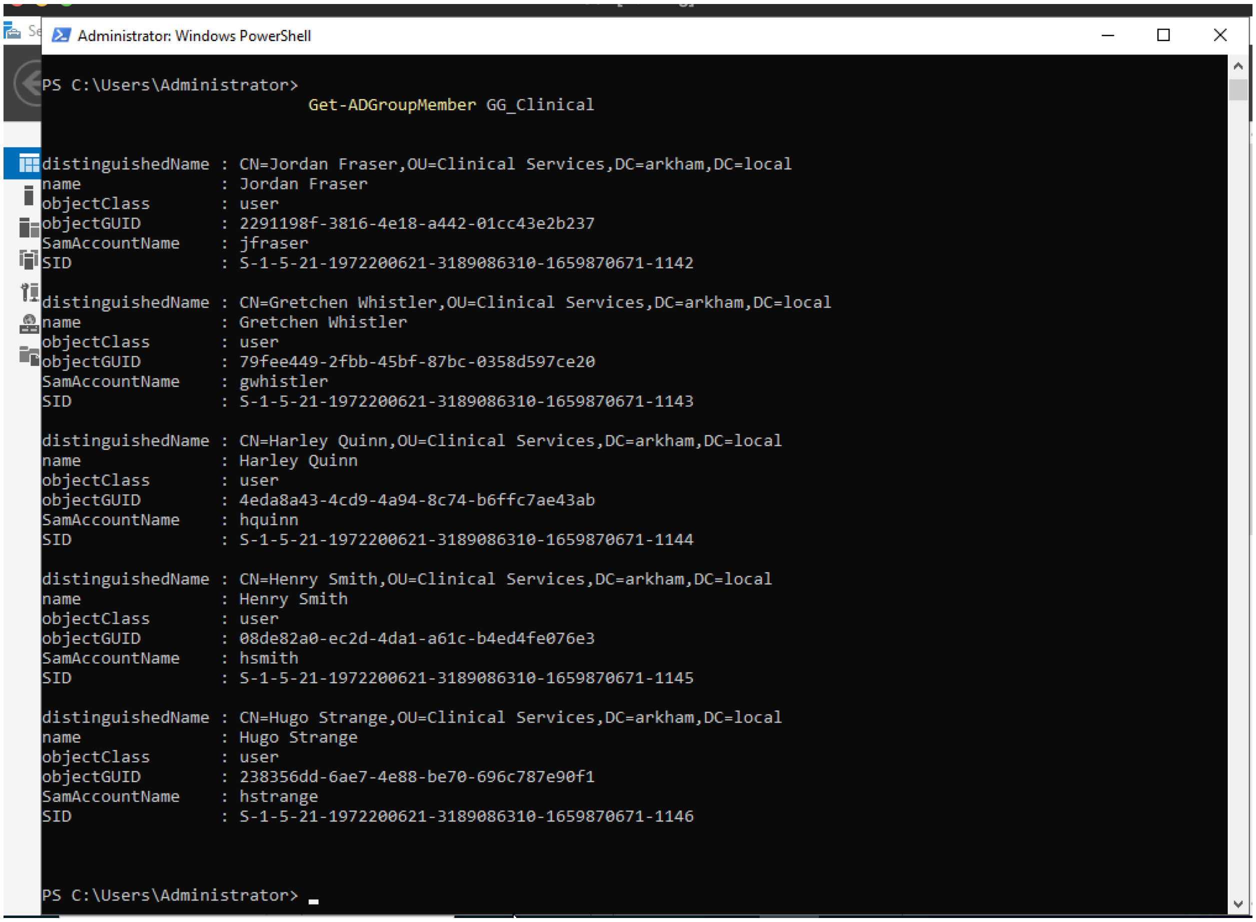
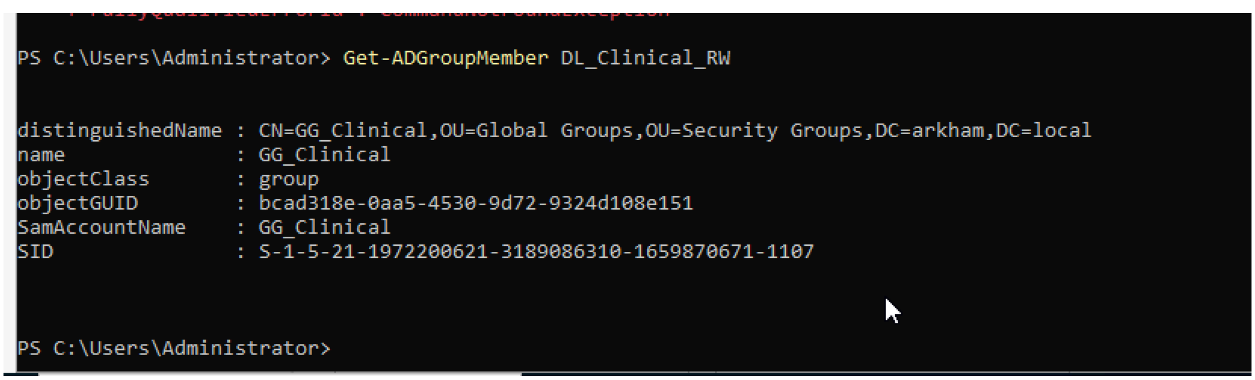


Figure 6.7.2 - Powershell was used to list all the users that are part of the GG_Clincal Security Group

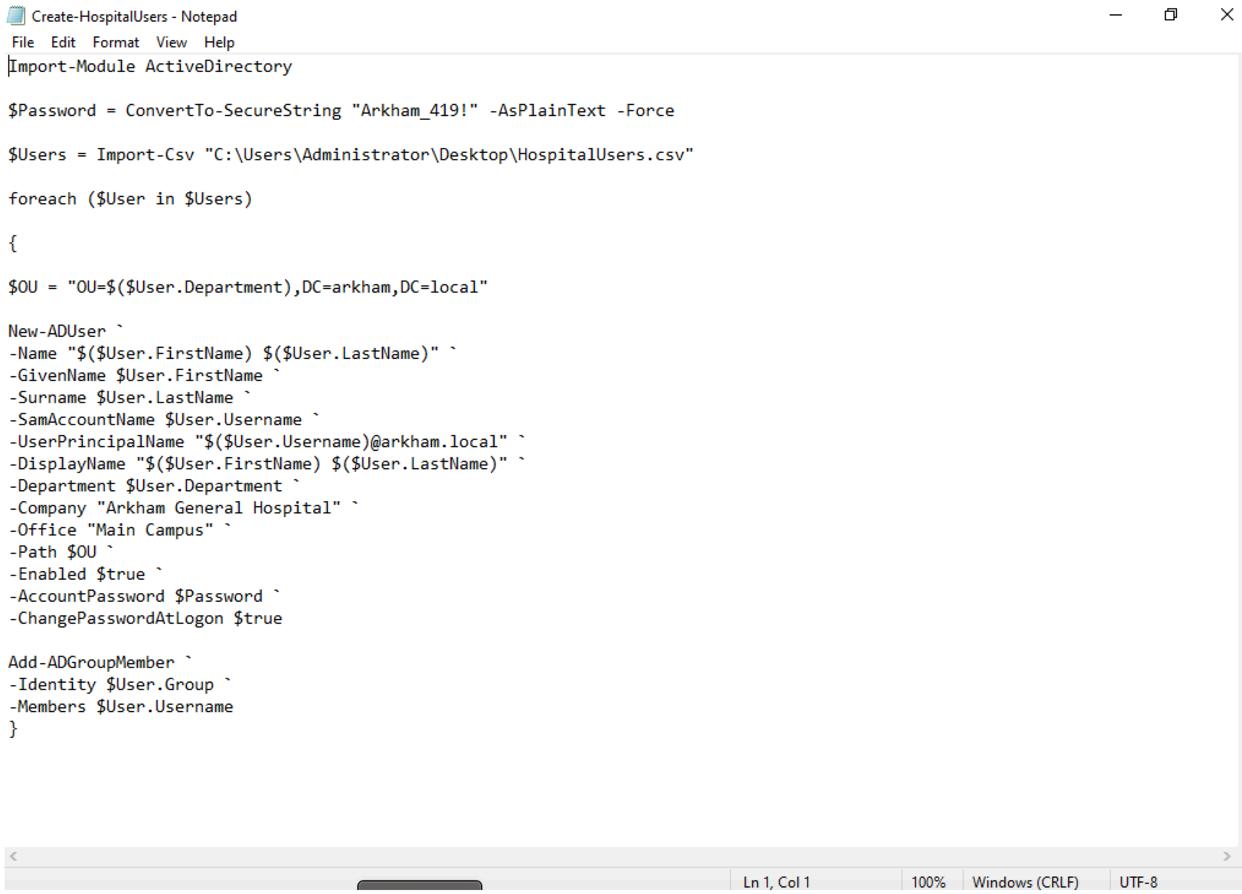




2. Powershell Script used to create multiple users simultaneously


```
FirstName, LastName, Username, Department, Group
Alfred, Pennyworth, apennyworth, Administration, GG_Administration
Aaron, Cash, acash,  Administration, GG_Administration
Adrian, Chen, achen, Administration, GG_Administration
Bruce, Wayne, bwayne, Administration, GG_Administration
Penelope, Young, pyoung, IT, GG_IT
Poison, Ivy, pivy, IT, GG_IT
Quincy, Sharp, qsharp, IT, GG_HelpDesk
Frank, Boles, fboles, IT, GG_ServerAdmins
Zach, Franklin, zfranklin, IT, GG_NetworkAdmins
Jordan, Fraser, jfraser, Clinical Services, GG_Clinical
Gretchen, Whistler, gwhistler, Clinical Services, GG_Clinical
Harley, Quinn, hquinn, Clinical Services, GG_Clinical
Henry, Smith, hsmith, Clinical Services, GG_Clinical
Hugo, Strange, hstrange, Clinical Services, GG_Clinical
Ian, Kennedy, ikennedy, Emergency, GG_Emergency
Jack, Ryder, jryder, Emergency, GG_Emergency
James, Gordon, jgordon, Emergency, GG_Emergency
Joe, Chill, jchill, Emergency, GG_Emergency
Carl, Todd, ctodd, Emergency, GG_Emergency
Killer, Croc, kcroc, Research, GG_Research
Thomas, Armbuster, tarmbuster, Research, GG_Research
Stephen, Kellerman, skellerman, Research, GG_Research
Sarah, Cassidy, scassidy, Research, GG_Research
Robert, Stirling, rstirling, Research, GG_Research
```

Figure 6.7.3 - LIst of users to be created

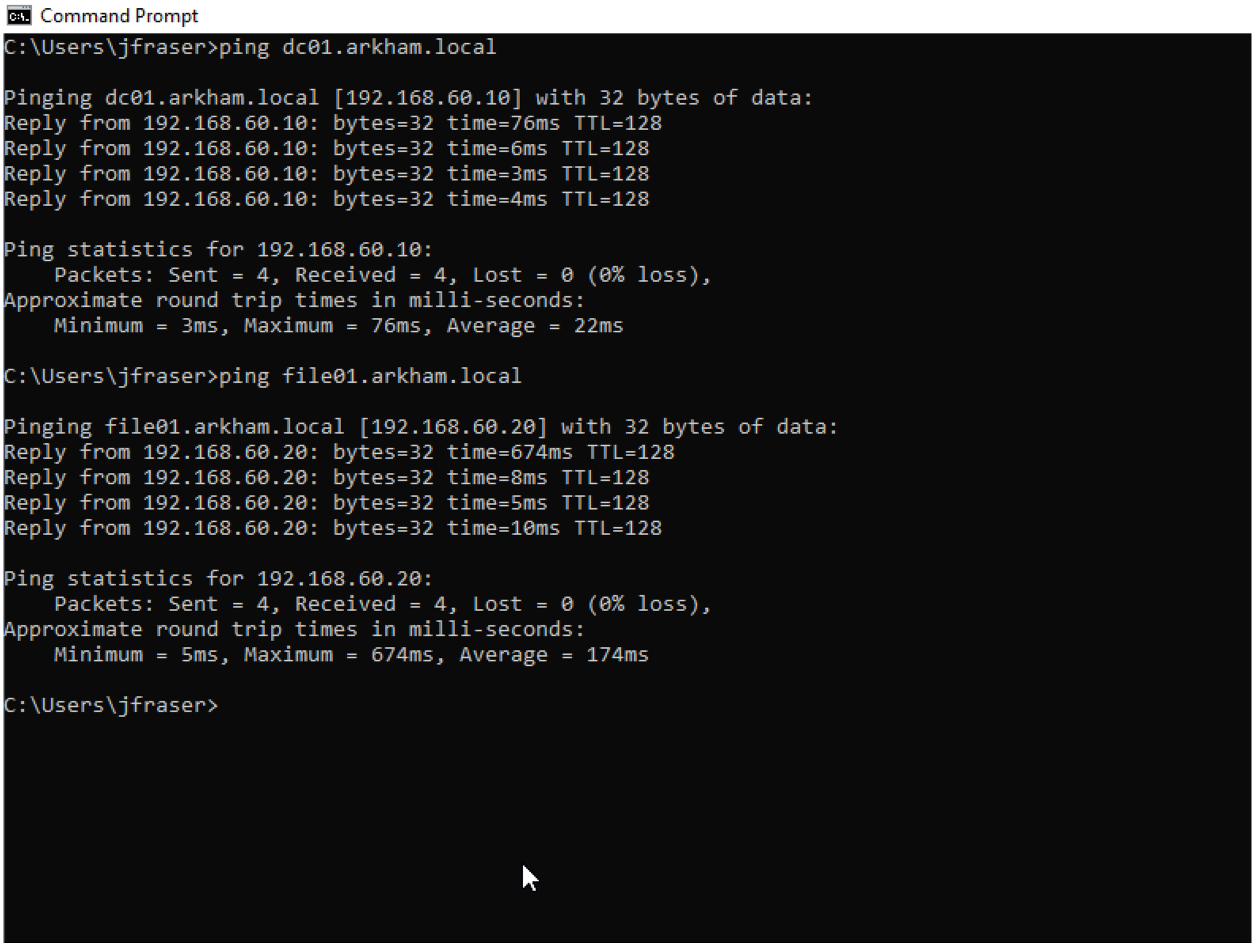


Create-HospitalUsers - Notepad

```
Import-Module ActiveDirectory

$Password = ConvertTo-SecureString "Arkham_419!" -AsPlainText -Force

$Users = Import-Csv "C:\Users\Administrator\Desktop\HospitalUsers.csv"

foreach ($User in $Users)

{

$OU = "OU=$($User.Department),DC=arkham,DC=local"

New-ADUser `
-Name "$($User.FirstName) $($User.LastName)" `
-GivenName $User.FirstName `
-Surname $User.LastName `
-SamAccountName $User.Username `
-UserPrincipalName "$($User.Username)@arkham.local" `
-DisplayName "$($User.FirstName) $($User.LastName)" `
-Department $User.Department `
-Company "Arkham General Hospital" `
-Office "Main Campus" `
-Path $OU `
-Enabled $true `
-AccountPassword $Password `
-ChangePasswordAtLogon $true

Add-ADGroupMember `
-Identity $User.Group `
-Members $User.Username
}
```

Figure 6.7.4- *Powershell script that created the list of users and placed them in their correct OU*

6.8 DNS and connectivity Validation

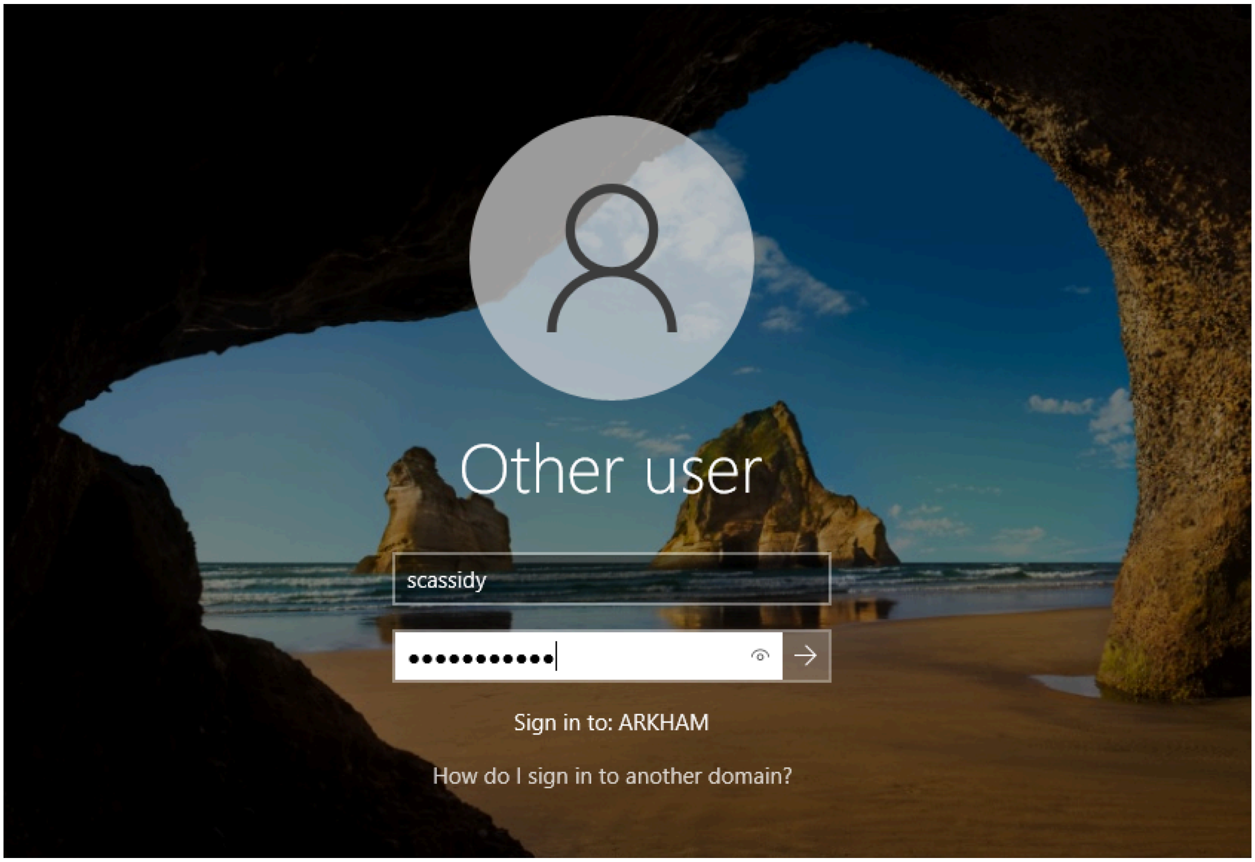


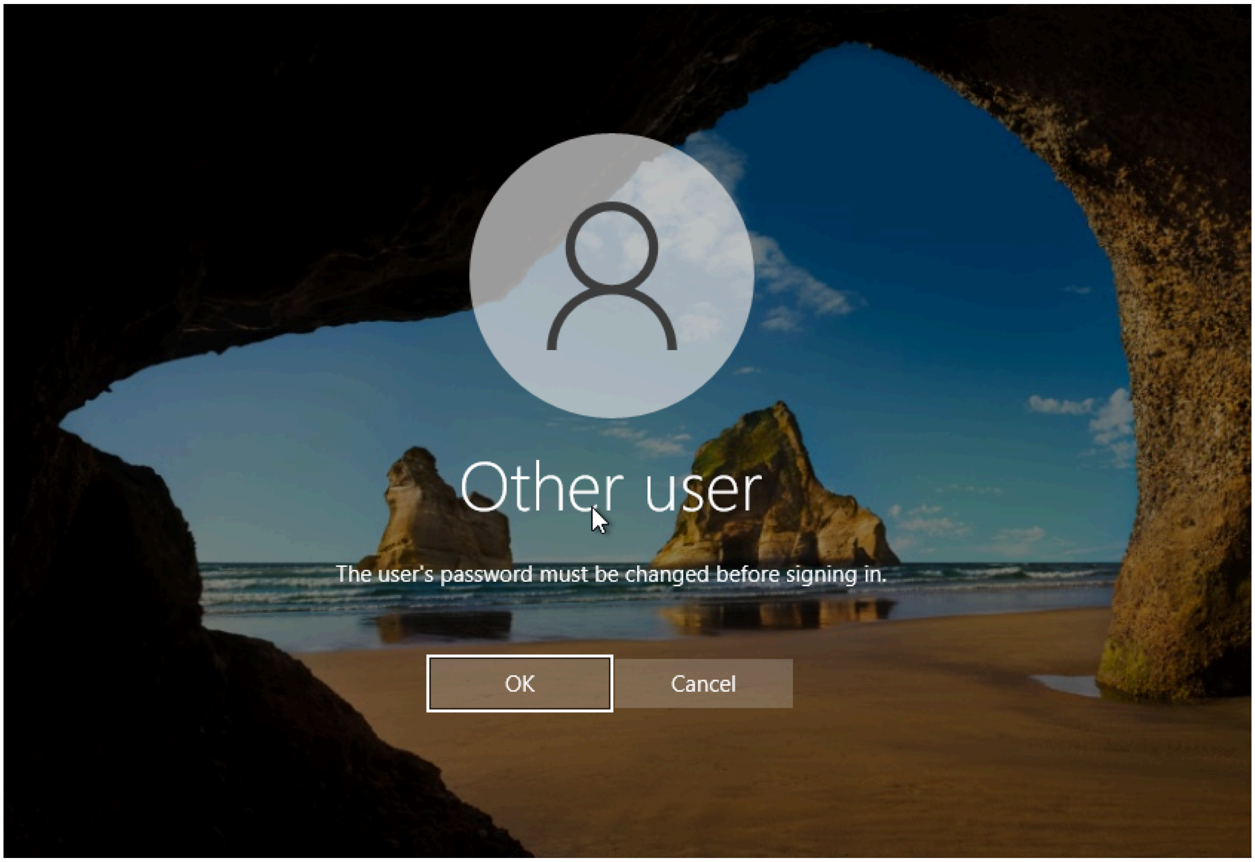
```
Command Prompt
C:\Users\jfraser>ping dc01.arkham.local

Pinging dc01.arkham.local [192.168.60.10] with 32 bytes of data:
Reply from 192.168.60.10: bytes=32 time=76ms TTL=128
Reply from 192.168.60.10: bytes=32 time=6ms TTL=128
Reply from 192.168.60.10: bytes=32 time=3ms TTL=128
Reply from 192.168.60.10: bytes=32 time=4ms TTL=128

Ping statistics for 192.168.60.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 3ms, Maximum = 76ms, Average = 22ms

C:\Users\jfraser>ping file01.arkham.local

Pinging file01.arkham.local [192.168.60.20] with 32 bytes of data:
Reply from 192.168.60.20: bytes=32 time=674ms TTL=128
Reply from 192.168.60.20: bytes=32 time=8ms TTL=128
Reply from 192.168.60.20: bytes=32 time=5ms TTL=128
Reply from 192.168.60.20: bytes=32 time=10ms TTL=128

Ping statistics for 192.168.60.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 5ms, Maximum = 674ms, Average = 174ms

C:\Users\jfraser>
```

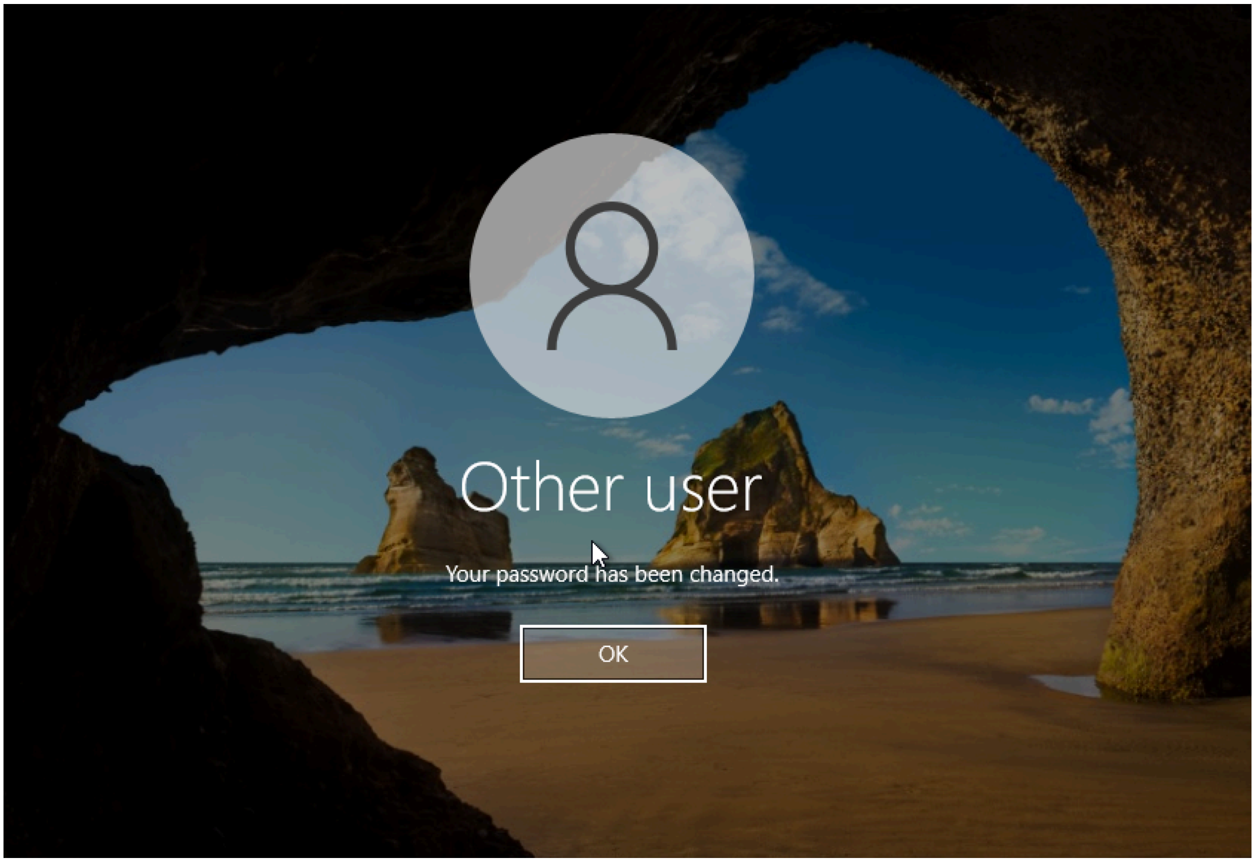
Forward DNS resolution successfully translates the hostnames dc01.arkham.local and file01.arkham.local to their corresponding IP addresses. Successful ICMP responses with 0% packet loss confirmed network connectivity from the domain-joined workstation to both infrastructure servers.

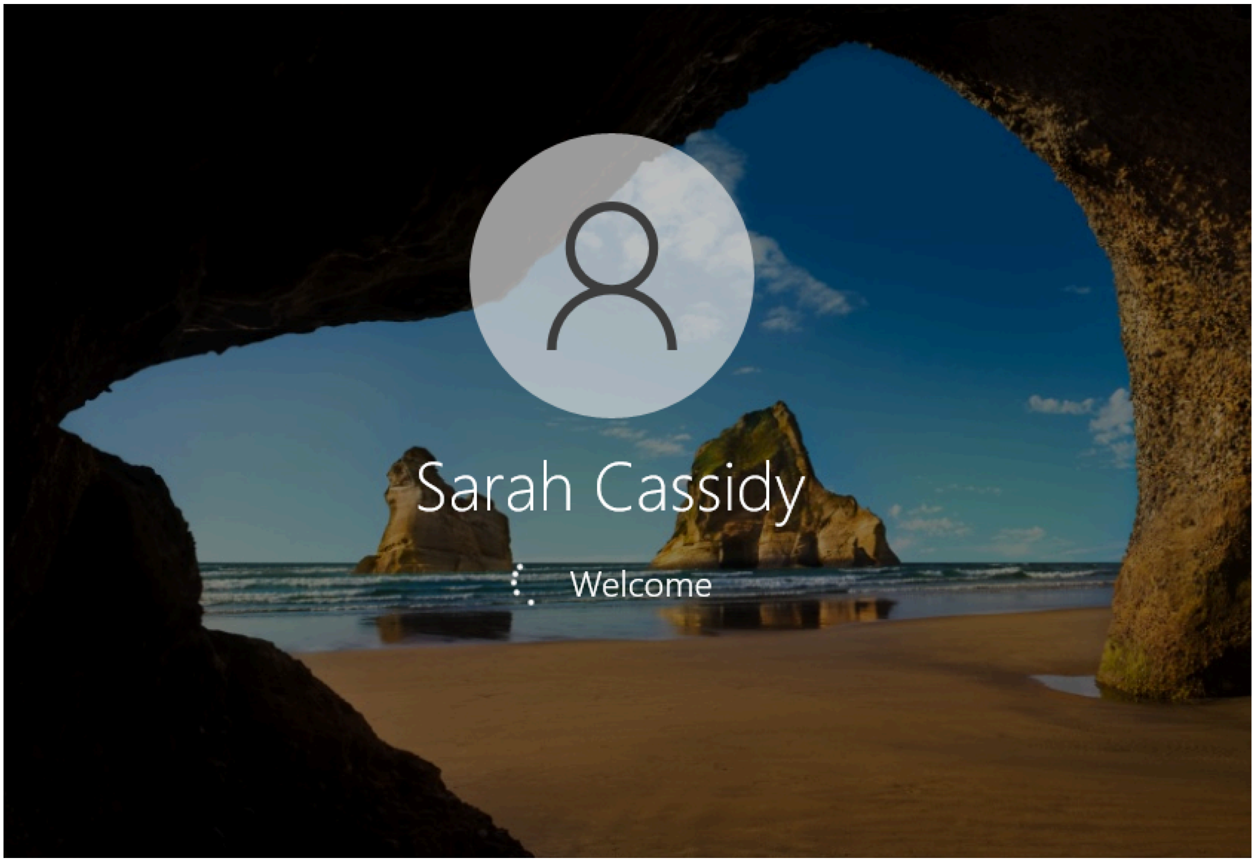
Test	Expected Result	Result
Dc01.arkham.local resolution	192.168.60.10	Passed
File01.arkham.local resolution	192.168.60.20	Passed
Ping DC01 by hostname	0% packet loss	Passed
Ping FILE01 by hostname	0% packet loss	Passed

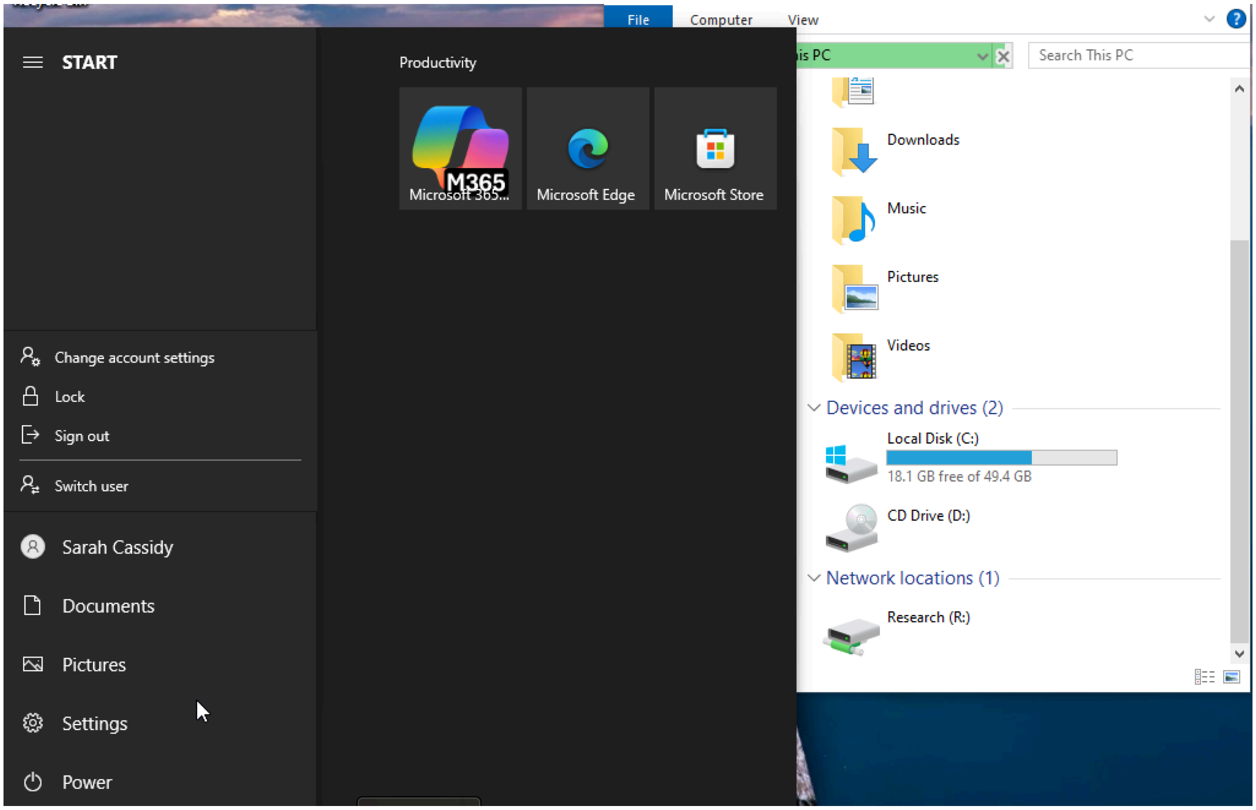
Figure 6.8.1 — Successful end-user validation of Research file-share access


Other user
scassidy
Sign in to: ARKHAM
How do I sign in to another domain?

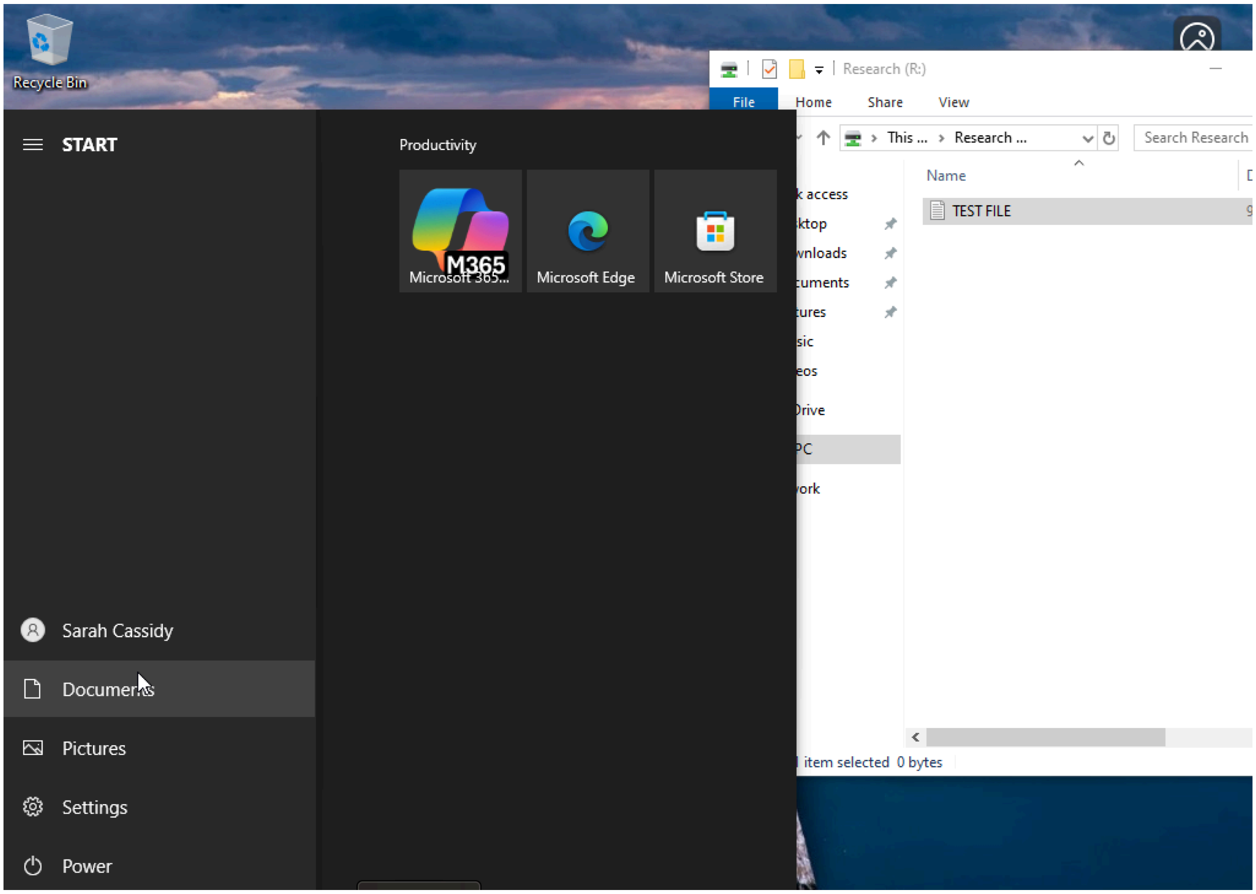

Other user
The user's password must be changed before signing in.
OK
Cancel


Other user
Your password has been changed.
OK


Sarah Cassidy
Welcome


File
Computer
View
is PC
Search This PC
START
Productivity
M365
Microsoft 365...
Microsoft Edge
Microsoft Store
Downloads
Music
Pictures
Videos
Devices and drives (2)
Local Disk (C:)
18.1 GB free of 49.4 GB
CD Drive (D:)
Network locations (1)
Research (R:)
Change account settings
Lock
Sign out
Switch user
Sarah Cassidy
Documents
Pictures
Settings
Power

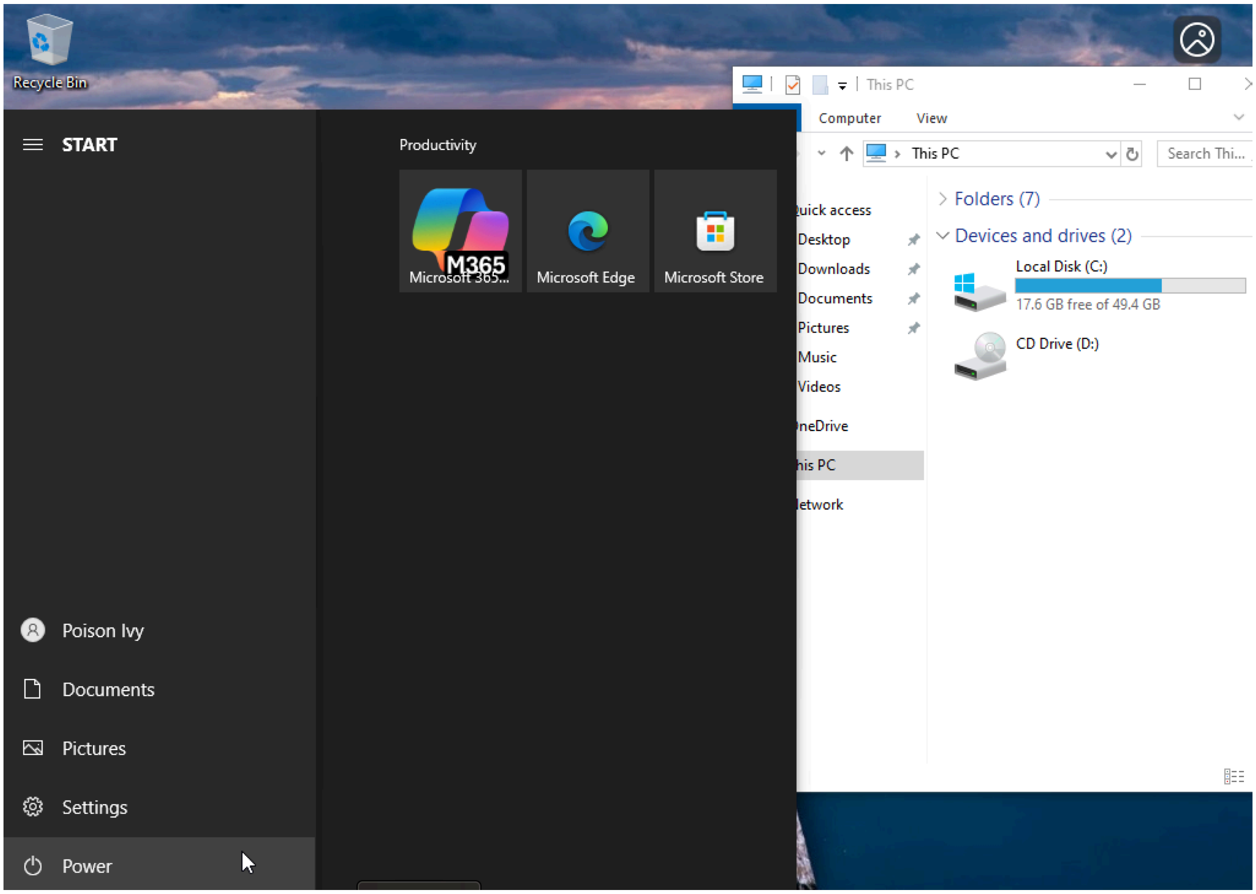
Confirmed that user can create a file and place it in the shared drive



6.9 End-User Troubleshooting

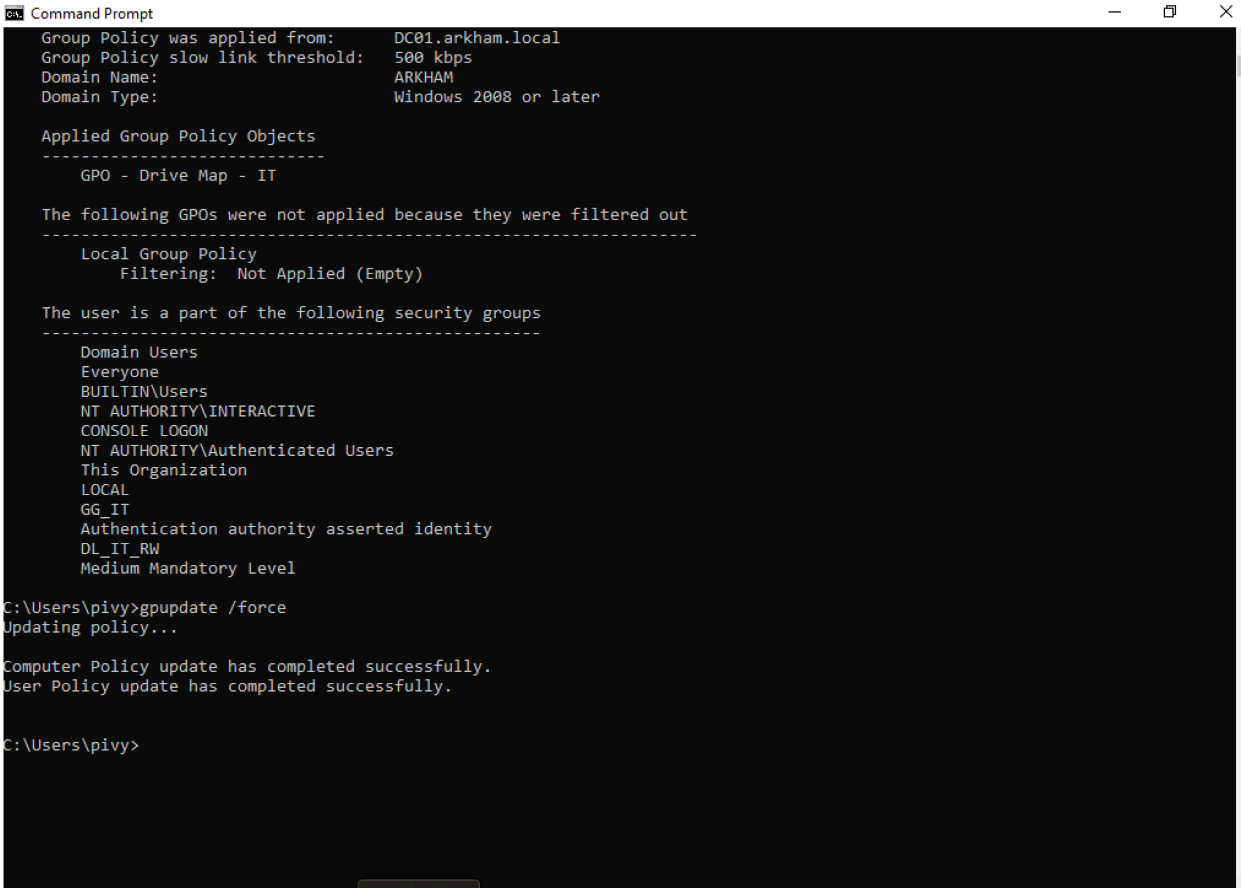
Incident #1 - Network Drive not Appearing

Figure 6.9.1 Confirm that network drive does not automatically appear when Poison Ivy signs in



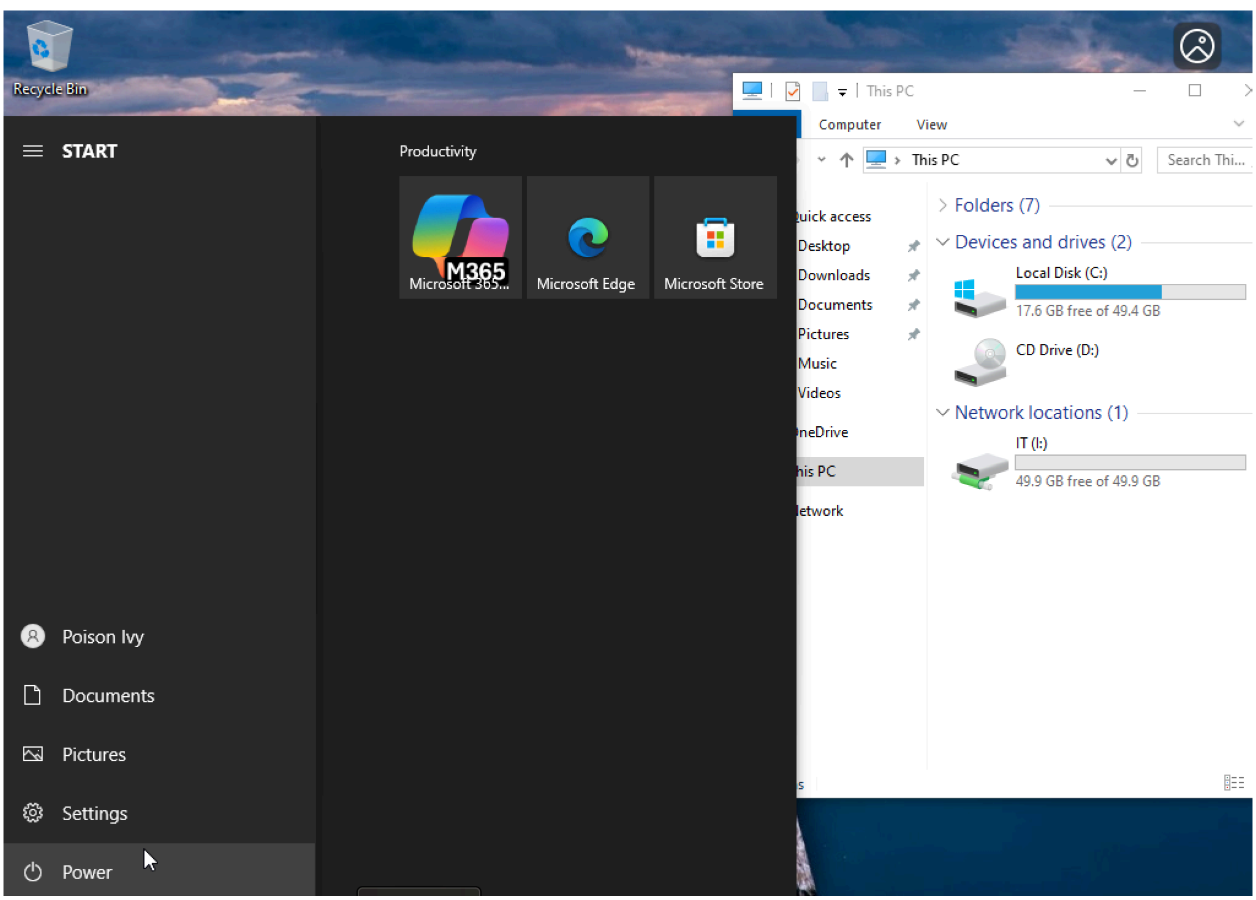
Initially ran gpresult /r to validate that the Applied Group Policy object was assigned to the user then ran gpupdate /force to push the policies onto the user's computer

Confirmed that user is now able to view shared drive:

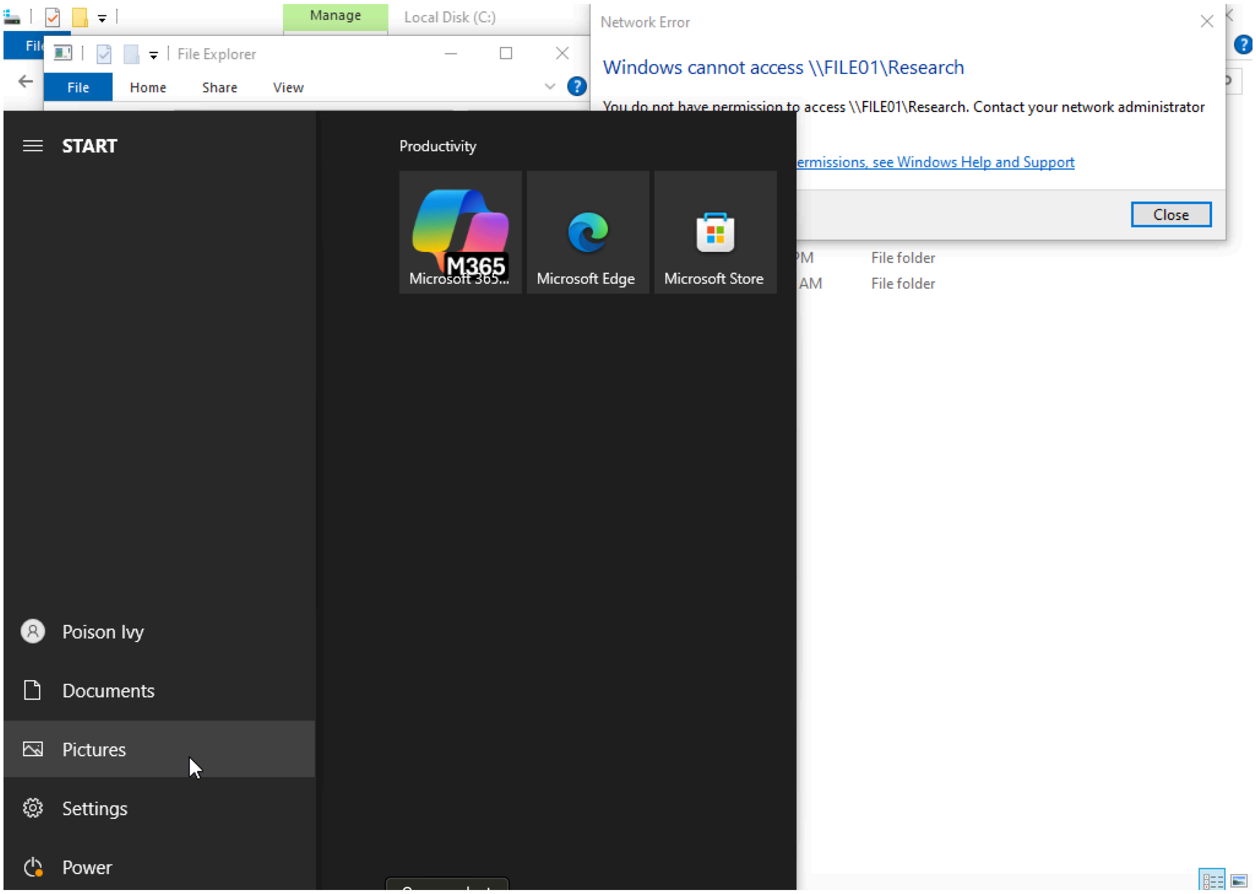

Command Prompt
Group Policy was applied from: DC01.arkham.local
Group Policy slow link threshold: 500 kbps
Domain Name: ARKHAM
Domain Type: Windows 2008 or later
Applied Group Policy Objects

GPO - Drive Map - IT
The following GPOs were not applied because they were filtered out
----
Local Group Policy
Filtering: Not Applied (Empty)
The user is a part of the following security groups

Domain Users
Everyone
BUILTIN\Users
NT AUTHORITY\INTERACTIVE
CONSOLE LOGON
NT AUTHORITY\Authenticated Users
This Organization
LOCAL
GG_IT
Authentication authority asserted identity
DL_IT_RW
Medium Mandatory Level
C:\Users\pivy>gpupdate /force
Updating policy...
Computer Policy update has completed successfully.
User Policy update has completed successfully.
C:\Users\pivy>


Recycle Bin
START
Productivity
M365
Microsoft 365...
Microsoft Edge
Microsoft Store
Poison Ivy
Documents
Pictures
Settings
Power
This PC
Computer
View
This PC
Search Thi...
Folders (7)
Devices and drives (2)
Local Disk (C:)
17.6 GB free of 49.4 GB
CD Drive (D:)
Network locations (1)
IT (I:)
49.9 GB free of 49.9 GB
Desktop
Downloads
Documents
Pictures
Music
Videos

Confirmed that Poison Ivy (IT user) cannot access the Research folder



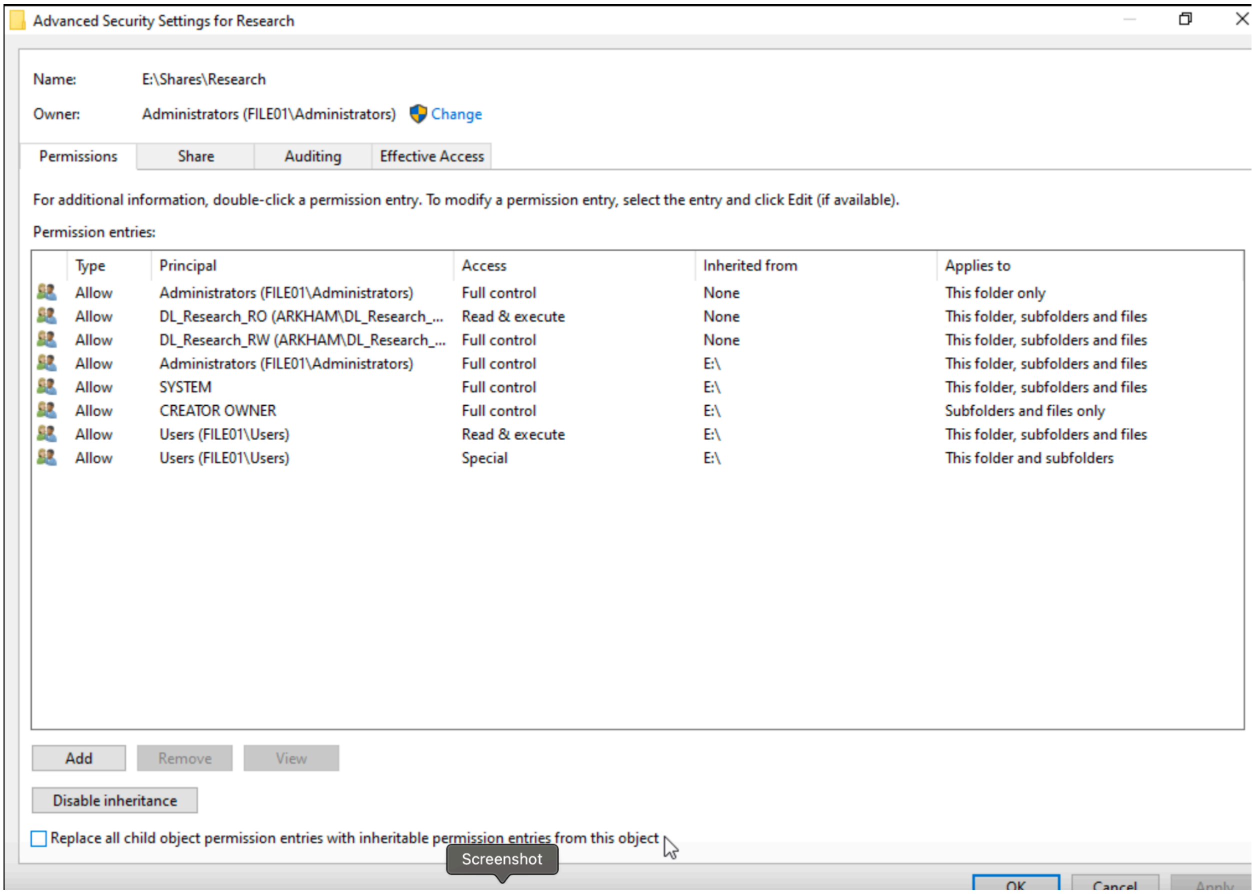
Cleanup of share permissions

Problem: Ordinary users are able to access the Research Folder

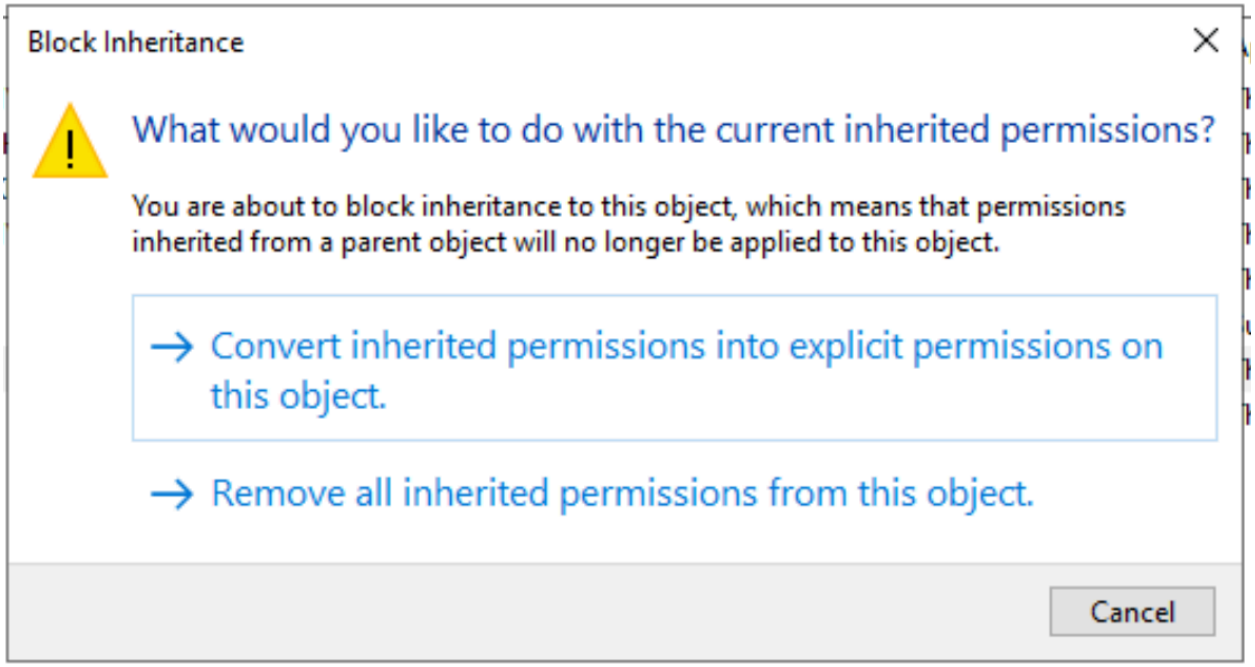
Evidence:

```
PS C:\Users\Administrator> (Get-Acl "E:\Shares\Research").Access | Select IdentityReference, FileSystemRights, AccessCotrolTy
pe, IsInherited

IdentityReference                 FileSystemRights AccessCotrolType IsInherited
-----------------                 ---------------- ---------------- -----------
BUILTIN\Administrators                 FullControl                        False
ARKHAM\DL_Research_RW                  FullControl                        False
ARKHAM\DL_Research_RO  ReadAndExecute, Synchronize                        False
BUILTIN\Administrators                 FullControl                         True
NT AUTHORITY\SYSTEM                    FullControl                         True
CREATOR OWNER                            268435456                         True
BUILTIN\Users          ReadAndExecute, Synchronize                         True
BUILTIN\Users                           AppendData                         True
BUILTIN\Users                          CreateFiles                         True
```

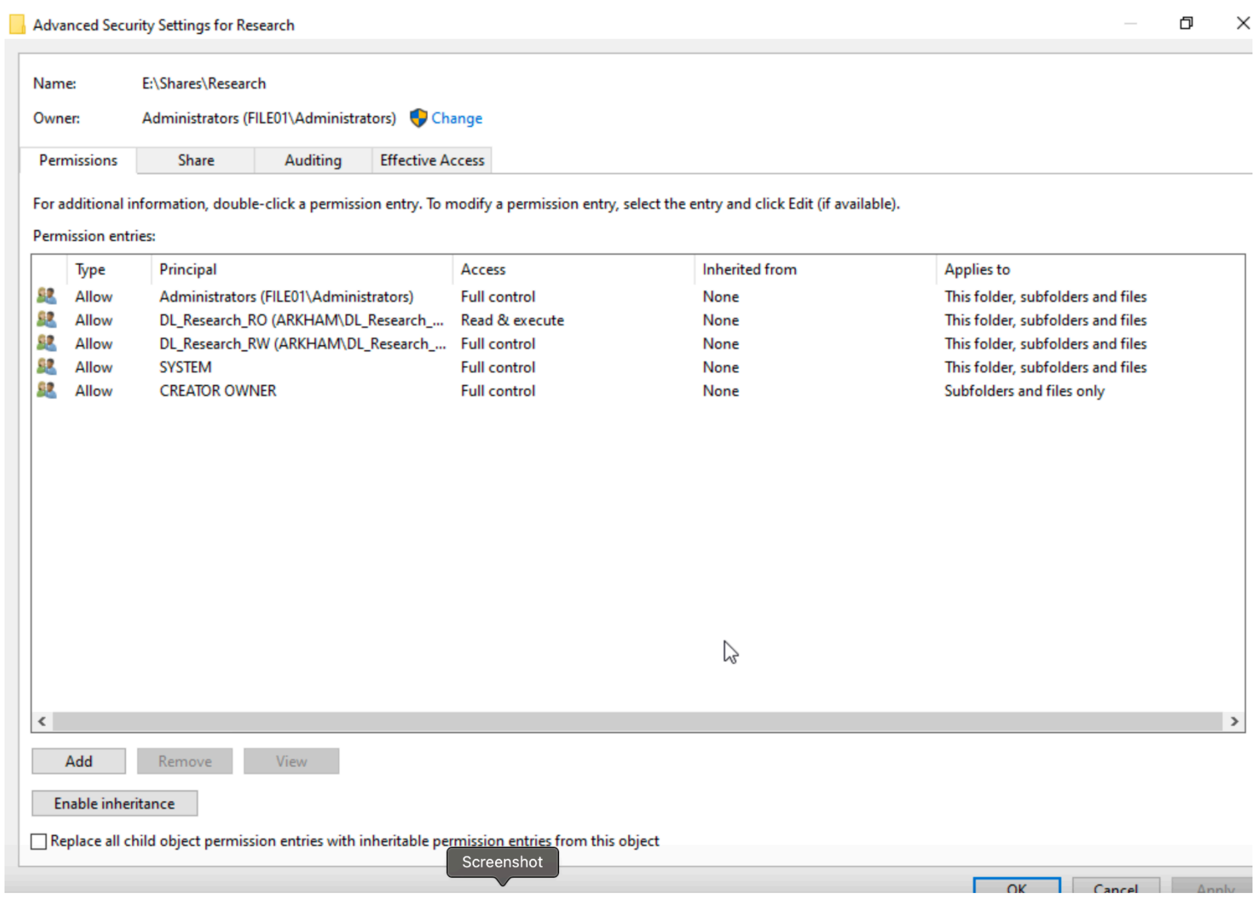
Advanced security settings for Research Folder which shows that ordinary users have read/write access:



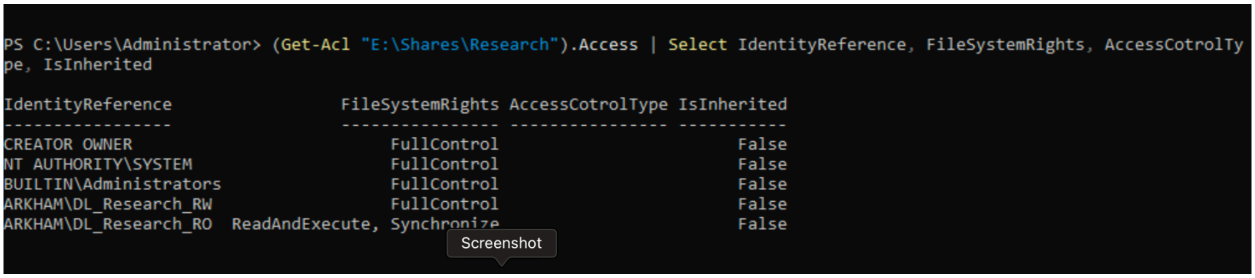
Remediation Step is to Convert Inherited permissions into explicit permissions


Block Inheritance
What would you like to do with the current inherited permissions?
You are about to block inheritance to this object, which means that permissions inherited from a parent object will no longer be applied to this object.
Convert inherited permissions into explicit permissions on this object.
Remove all inherited permissions from this object.
Cancel

Final

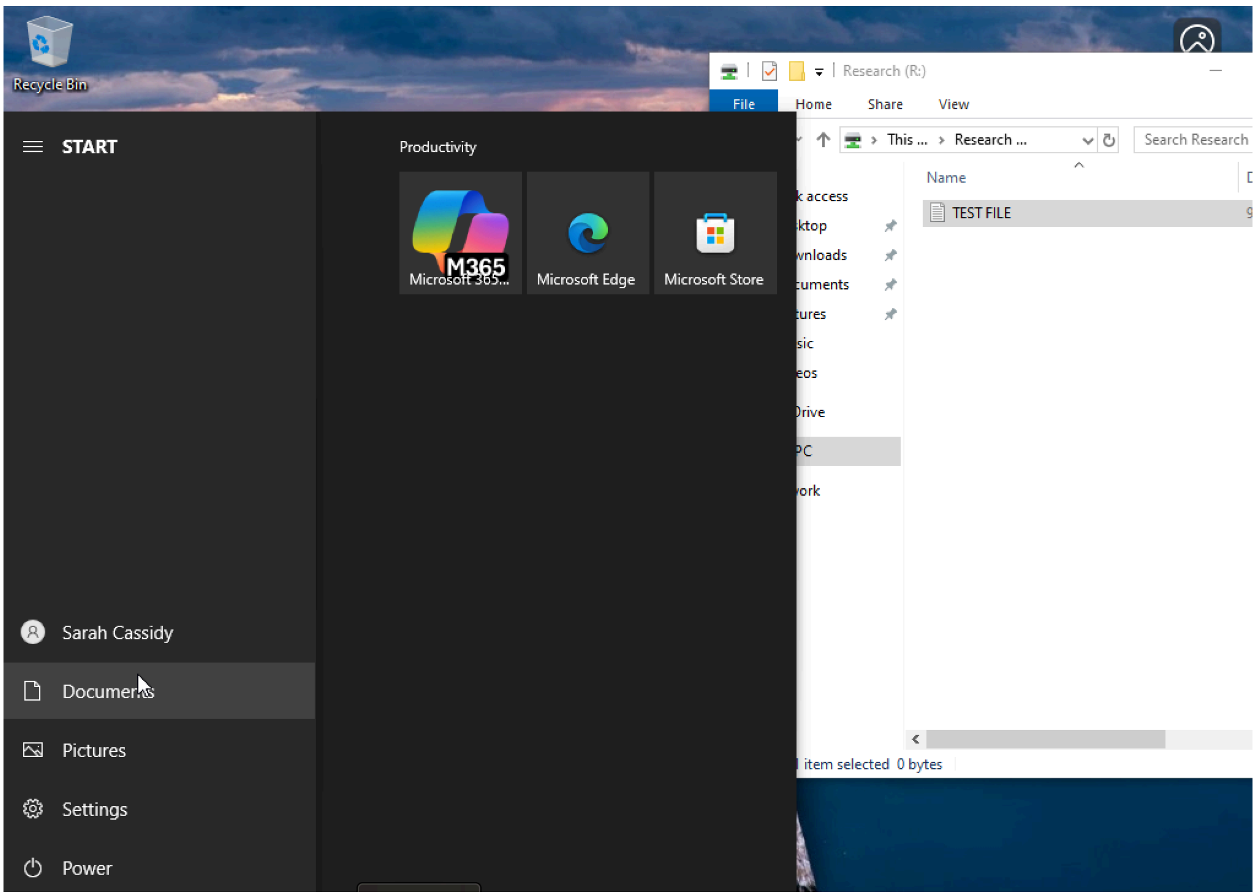

Advanced Security Settings for Research
Name: E:\Shares\Research
Owner: Administrators (FILE01\Administrators) Change
Permissions
Share
Auditing
Effective Access
For additional information, double-click a permission entry. To modify a permission entry, select the entry and click Edit (if available).
Permission entries:
Type Principal Access Inherited from Applies to
Allow Administrators (FILE01\Administrators) Full control None This folder, subfolders and files
Allow DL_Research_RO (ARKHAM\DL_Research_... Read & execute None This folder, subfolders and files
Allow DL_Research_RW (ARKHAM\DL_Research_... Full control None This folder, subfolders and files
Allow SYSTEM Full control None This folder, subfolders and files
Allow CREATOR OWNER Full control None Subfolders and files only
Add
Remove
View
Enable inheritance
Replace all child object permission entries with inheritable permission entries from this object
Screenshot
Cancel

NTFS permissions are explicitly assigned to the research read/write and read-only security groups, while generic user access has been removed

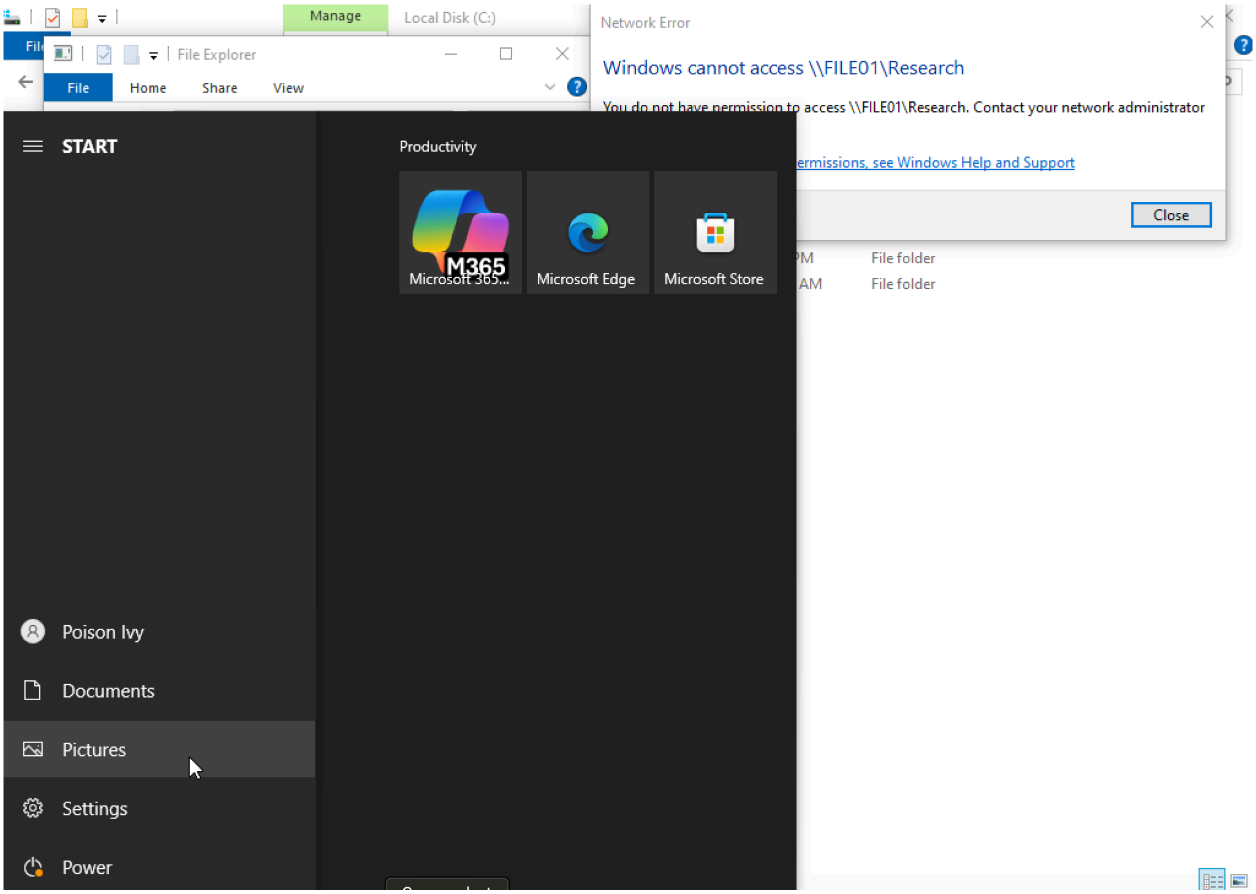


```
PS C:\Users\Administrator> (Get-Acl "E:\Shares\Research").Access | Select IdentityReference, FileSystemRights, AccessCotrolTy
pe, IsInherited

IdentityReference                  FileSystemRights AccessCotrolType IsInherited
-----------------                  ---------------- ---------------- -----------
CREATOR OWNER                           FullControl                        False
NT AUTHORITY\SYSTEM                     FullControl                        False
BUILTIN\Administrators                  FullControl                        False
ARKHAM\DL_Research_RW                   FullControl                        False
ARKHAM\DL_Research_RO  ReadAndExecute, Synchronize                         False
```

Proof that Research User can create file in Research folder:



Proof that non-research user cannot access the Research folder:


Manage
Local Disk (C:)
File Explorer
File
Home
Share
View
Network Error
Windows cannot access \\FILE01\Research
You do not have permission to access \\FILE01\Research. Contact your network administrator
ermissions, see Windows Help and Support
Close
START
Productivity
M365
Microsoft 365...
Microsoft Edge
Microsoft Store
File folder
File folder
Poison Ivy
Documents
Pictures
Settings
Power